

# **INTRODUCTION TO INFORMATION AND COMMUNICATION TECHNOLOGIES**

Q. Define ICT. Explain the advantages of ICT.

Ans. Definition:

ICT (Information and Communication Technology) is a term that describes all the tools and systems people use to share, store, and process information. This includes things like computers, the internet, phones, and software that help people communicate, work, and manage data. It is basically how we use technology to connect and handle information in our daily lives.

ICT refers to the integration of **computing, networking, and information processing technologies** to collect, process, store, and transmit data and information efficiently and securely for business, education, government, or personal purposes."

Advantages of ICT (Information and Communication Technology):

## **1. Enhanced Communication**

ICT enables quick and easy communication through emails, instant messaging, video calls, and social media. This helps people stay connected regardless of their location.

## **2. Access to Information**

With ICT, people can access vast amounts of information from anywhere via the internet. This helps with learning, research, and staying informed about global events.

## **3. Improved Education**

ICT supports learning through online courses, educational software, and e-books. It makes education more interactive and accessible, allowing for remote learning and a wider reach to students.

## **4. Increased Efficiency and Productivity**

Technology helps automate tasks, streamline operations, and improve workflow. Businesses can use software to manage projects, track progress, and communicate effectively, saving time and reducing errors.

## **5. Economic Growth**

ICT has boosted economic activities by enabling e-commerce, online businesses, and digital transactions. It has opened up new markets and job opportunities.

## **6. Innovation and Creativity**

ICT fosters innovation by providing the tools and platforms for developing new technologies, applications, and creative projects. This leads to advancements in various fields, including healthcare and engineering.

**7. Remote Work Opportunities**

ICT allows people to work from home or any location with an internet connection. This flexibility can improve work-life balance and expand job options.

**8. Better Services in Healthcare**

ICT supports telemedicine, electronic health records, and remote patient monitoring, improving patient care by making it easier to share and access medical information.

**9. Enhanced Collaboration**

People can collaborate in real-time on projects from different parts of the world using ICT tools such as cloud services, shared documents, and communication platforms.

**10. Global Connectivity**

ICT has broken down geographical barriers, enabling people and businesses to interact globally, which fosters cultural exchange and international collaboration.

These advantages show how ICT has become an essential part of modern society, contributing to greater convenience, growth, and connectedness.

**Q: Write in detail the components of ICT.**

**Ans. Components of ICT**

These are the components

**1. Computers and Computing Technology**

- **Hardware:** Physical components such as desktops, laptops, servers, storage devices, and input/output devices.
- **Software:** Programs and applications that run on computers, enabling users to perform specific tasks. These include operating systems, productivity tools, and specialized software for various industries.

**2. Telecommunications**

- **Telecommunication Networks:** Systems that allow for data transfer and communication over distances, including telephone lines, mobile networks, and satellite communication.
- **Internet Infrastructure:** The global network that connects millions of private, public, academic, business, and government networks. It includes routers, servers, and data centers.

**3. Networking Components**

- **LAN (Local Area Network), MAN and WAN (Wide Area Network):** Networks that connect devices within a limited area (LAN) or over broader geographical areas (WAN).
- **Wi-Fi and Wireless Networks:** Technologies that enable wireless data transfer, providing connectivity without the need for cables.

**4. Data Management and Storage**

- **Cloud Computing:** A model that allows for the delivery of computing services over the internet, including data storage, databases, servers, and software.
- **Databases:** Systems that store and organize large sets of data, making it accessible and manageable for organizations.

## 5. Media and Content

**Digital Media** Encompasses the creation, storage, and transmission of multimedia content like videos, audio, and images. Platforms include video hosting services, social media, and online streaming.

**Content Management Systems (CMS)** Software tools that facilitate the creation, management, and modification of digital content.

## 6. Communication Tools

• **Email and Messaging Services** Platforms for exchanging messages over the internet, including Gmail, Outlook, and instant messaging services.

• **Video Conferencing and Collaboration Tools** Applications like Zoom, Microsoft Teams, and Skype that allow real-time video communication and collaboration.

## 7. Security and Privacy Components

• **Cybersecurity Measures** Tools and practices designed to protect networks, devices, programs, and data from cyberattacks. This includes firewalls, antivirus software, and encryption techniques.

• **Data Protection Policies** Legal and operational guidelines to ensure user privacy and data protection, such as the General Data Protection Regulation (GDPR).

## 8. Internet of Things (IoT)

• **Smart Devices and Automation** The integration of physical devices connected to the internet, enabling them to send and receive data. This includes smart home systems, wearable technology, and connected vehicles.

## 9. Artificial Intelligence (AI) and Machine Learning (ML)

• **AI Applications** Systems that simulate human intelligence processes such as learning, reasoning, and problem-solving. These are applied in areas like chatbots, recommendation systems, and robotics.

• **ML Algorithms** A subset of AI where systems learn and make predictions based on data analysis without being explicitly programmed.

## 10. Support and Maintenance

• **IT Support Services** Teams and systems dedicated to maintaining ICT infrastructure, troubleshooting issues, and ensuring continuous operation.

• **Training and Education** Programs to equip users with the skills needed to effectively use ICT tools.

**Q:** Write in detail about the scope of ICT?

**Ans.** Scope of ICT

Information and Communication Technology (ICT) plays a vital role in shaping modern society by transforming how we learn, work, govern, receive healthcare, and entertain ourselves. Its applications span across every major field, bringing efficiency, innovation, and global connectivity.

### 1. ICT in Education

ICT has revolutionized the education sector by enabling digital learning, online classes, and access to a vast range of resources beyond traditional textbooks. Tools like Learning Management Systems (LMS), interactive whiteboards, and virtual classrooms promote

personalized learning experiences. ICT also supports distance learning, making education accessible to remote and underserved areas.

## 2. ICT in Business

In the business world, ICT enhances communication, automates processes, and facilitates global operations. From enterprise resource planning (ERP) systems and customer relationship management (CRM) tools to e-commerce platforms and cloud services, ICT helps businesses improve efficiency, decision-making, and customer engagement. It also enables remote work and global collaboration.

## 3. ICT in Governance

Governments use ICT to improve transparency, deliver public services efficiently, and promote citizen participation. This includes e-governance platforms, online portals for documentation and tax filing, and digital voting systems. ICT helps streamline administrative processes and makes governance more accessible and accountable to the public.

## 4. ICT in Healthcare

ICT supports the healthcare industry through electronic health records (EHRs), telemedicine, health information systems, and wearable health devices. It allows for remote diagnosis, better data management, and improved patient care. ICT also facilitates research, medical education, and real-time monitoring of patients using IoT-enabled devices.

## 5. ICT in Digital Media and Entertainment

ICT is the backbone of the digital media and entertainment industry. Streaming services, video games, social media platforms, and digital content creation rely heavily on ICT tools and infrastructure. It enables content distribution on a global scale, personalized viewing experiences, and real-time audience interaction.

The scope of ICT extends into every aspect of modern life, continuously evolving and expanding its influence. From education to entertainment, ICT enhances accessibility, efficiency, and innovation, making it a cornerstone of development in the digital age.

**Q: Write in detail on the tools of ICT.**

**Ans: ICT Tools: Enhancing Communication, Collaboration, and Productivity**

Information and Communication Technology (ICT) tools are the instruments, software applications, and digital resources used to collect, process, store, and communicate information. These tools play a vital role in modern-day education, business, healthcare, governance, and day-to-day communication. They have transformed the way we work, learn, and interact globally.

### Categories of ICT Tools

ICT tools can be classified based on their primary function and domain of usage:

#### 1. Communication Tools

These tools allow people to share information and communicate in real time or asynchronously.

- **Email Tools:** Gmail, Outlook
- **Instant Messaging:** WhatsApp, Messenger, Telegram
- **Video Conferencing Tools:** Zoom, Google Meet, Microsoft Teams
- **Voice over IP (VoIP):** Skype, Viber

## 2. Collaboration Tools

These tools enable teamwork and shared productivity across locations.

- **Project Management:** Trello, Asana, Monday.com
- **Team Communication:** Slack, Microsoft Teams
- **Collaborative Editing:** Google Docs, Google Sheets, Office 365
- **Shared Storage:** Google Drive, Dropbox, OneDrive

## 3. Productivity Tools

These tools are used to create documents, presentations, spreadsheets, and manage tasks.

- **Word Processing:** Microsoft Word, Google Docs
- **Spreadsheets:** Microsoft Excel, Google Sheets
- **Presentation Tools:** Microsoft PowerPoint, Prezi, Canva
- **Note-Taking Tools:** Evernote, OneNote, Notion

## 4. Teaching and Learning Tools

These tools are used in educational settings for content delivery, assessment, and interaction.

- **Learning Management Systems (LMS):** Moodle, Edmodo, Google Classroom
- **Quiz and Assessment Tools:** Kahoot, Quizizz, Socrative
- **Virtual Labs and Simulators:** PhET Interactive Simulations, Labster
- **E-Learning Platforms:** Coursera, Udemy, Khan Academy

## 5. Data Management Tools

These tools are used to collect, organize, analyze, and visualize data.

- **Database Management Systems (DBMS):** MySQL, Oracle, MS Access
- **Data Analysis Tools:** Microsoft Power BI, Tableau, Google Data Studio
- **Survey Tools:** Google Forms, SurveyMonkey, Microsoft Forms

## 6. Graphic Design and Multimedia Tools

These tools are used for image, video, and audio creation/editing.

- **Graphic Design:** Adobe Photoshop, Illustrator, Canva
- **Video Editing:** Adobe Premiere Pro, Camtasia, iMovie
- **Audio Editing:** Audacity, GarageBand
- **Presentation Design:** Canva, Prezi

## 7. Web Development Tools

These tools are used to create and maintain websites and online services.

- **Code Editors:** VS Code, Sublime Text
- **Web Frameworks:** WordPress, Bootstrap
- **Website Builders:** Wix, Squarespace
- **FTP Clients:** FileZilla

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**8. Security Tools**

These tools are used to protect digital data and ensure safe ICT practices.

- **Antivirus Software:** Norton, Avast, Kaspersky
- **Firewall Tools:** Windows Firewall, ZoneAlarm
- **Encryption Tools:** VeraCrypt, BitLocker
- **Authentication Tools:** Google Authenticator, Duo Security

**Benefits of ICT Tools****1. Increased Efficiency**

ICT tools help automate repetitive tasks and streamline workflows, which significantly boosts productivity. They also allow users to manage and process large volumes of data with speed and accuracy.

**2. Improved Communication**

With ICT tools like email, messaging apps, and video conferencing, communication becomes instant and seamless. People can connect across long distances in real time, breaking down geographical barriers.

**3. Enhanced Collaboration**

ICT platforms support real-time collaboration, enabling multiple users to work on the same project simultaneously. This fosters teamwork, improves coordination, and accelerates project completion.

**4. Remote Access**

One of the major benefits of ICT tools is the ability to work and learn from anywhere. As long as there is internet access, users can connect to systems and resources remotely.

**5. Cost Savings**

By reducing the need for physical infrastructure, travel, and printed materials, ICT tools help organizations and individuals save money. Virtual meetings and cloud-based systems minimize operational costs.

**Q: Write in detail note on emerging trends in ICT Platforms.**

**Ans. Emerging Trends in ICT Platforms**

As Information and Communication Technologies (ICT) evolve, several cutting-edge innovations are transforming how platforms operate, communicate, and deliver services. The following are some of the most influential emerging trends in ICT platforms:

**1. Artificial Intelligence (AI) Integration**

Artificial Intelligence is increasingly being embedded into ICT platforms to enhance decision-making, automate tasks, and improve user experiences. AI algorithms enable systems to learn from data, predict outcomes, and adapt to new inputs. Applications include chatbots in customer support, recommendation systems in e-commerce, and intelligent personal assistants like Siri or Google Assistant. AI also plays a crucial role in data analytics, cybersecurity, and adaptive learning systems.

## 2. Internet of Things (IoT)

The Internet of Things connects everyday physical objects—such as appliances, vehicles, and wearable devices—to the internet, enabling them to collect and exchange data. ICT platforms integrated with IoT are used in smart homes, smart cities, agriculture, and industrial automation. These systems can monitor environments, detect changes, and take action in real time, leading to more efficient resource management and improved quality of life.

## 3. Edge Computing

Edge computing refers to processing data closer to the source of generation (the "edge" of the network), rather than relying solely on centralized cloud servers. This reduces latency, enhances real-time processing, and lowers bandwidth usage. In ICT platforms, edge computing is particularly useful for time-sensitive applications such as autonomous vehicles, healthcare monitoring, and industrial IoT systems. It complements cloud computing by distributing computing power across a wider network.

## 4. Blockchain for Secure Transactions

Blockchain is a decentralized, distributed ledger technology that ensures secure and transparent record-keeping. In ICT platforms, blockchain is being used to enhance data security, facilitate secure financial transactions, and manage digital identities. Its application extends to supply chain management, voting systems, smart contracts, and health records. By eliminating the need for intermediaries, blockchain increases trust, reduces fraud, and improves efficiency.

## 5. Virtual and Augmented Reality (VR/AR)

VR and AR technologies are revolutionizing user interaction with ICT platforms. Virtual Reality immerses users in a completely digital environment, while Augmented Reality overlays digital content on the real world. These technologies are used in education for immersive learning, in healthcare for training and surgery simulations, and in e-commerce for virtual try-ons. As VR/AR hardware becomes more affordable, their integration into ICT platforms is expected to grow rapidly.

## 6. Quantum Computing

Quantum computing represents a breakthrough in computational power, using the principles of quantum mechanics to perform calculations at unimaginable speeds. While still in its early stages, its integration into ICT platforms promises revolutionary impacts in cryptography, complex simulations, artificial intelligence, and data processing. Quantum computers can solve problems that are currently intractable for classical computers.

## 7. 5G Technology and Beyond

The rollout of 5G technology is setting a new standard for wireless communication, providing ultra-fast internet speeds, low latency, and massive device connectivity. ICT platforms powered by 5G support real-time applications such as autonomous vehicles, remote surgery, and advanced mobile experiences. Looking forward, **6G and beyond** aim to bring even more intelligent, AI-driven, and seamless connectivity across the globe.

The integration of AI, IoT, edge computing, blockchain, and VR/AR into ICT platforms marks a significant shift toward smarter, more efficient, and more secure systems. These emerging trends not only enhance the capabilities of existing platforms but also pave the way for entirely new applications and services across various sectors.

**Short Questions**

1. **What is ICT?**  
Ans. ICT stands for Information and Communication Technology, encompassing all technologies used to handle information and enable communication.
2. **Why is ICT important in modern life?**  
Ans. ICT connects people globally, supports education, business, healthcare, and improves access to information.
3. **What are the fundamental concepts of ICT?**  
Ans. The key concepts include data processing, communication, hardware, software, and networks.
4. **What are examples of hardware in ICT?**  
Ans. Hardware includes input devices (keyboard, mouse), output devices (monitor, printer), storage devices, and processors.
5. **What is system software?**  
Ans. System software manages hardware and provides a platform for running application software, e.g., operating systems.
6. **What is application software?**  
Ans. Application software performs specific tasks for users like word processors, web browsers, and games.
7. **What are ICT platforms?**  
Ans. ICT platforms are tools and technologies such as cloud services, databases, and development environments used in ICT.
8. **What is the difference between LAN and WAN?**  
Ans. LAN (Local Area Network) covers a small area like a home or office; WAN (Wide Area Network) connects larger areas, such as cities or countries.
9. **What is the Internet?**  
Ans. The Internet is a global network connecting millions of private, public, academic, business, and government networks.
10. **What are the types of data storage in ICT?**  
Ans. Data can be stored locally on hard drives or remotely in cloud-based storage systems.
11. **How is ICT used in education?**  
Ans. ICT supports digital classrooms, e-learning platforms, and access to vast educational resources.
12. **What role does ICT play in business?**  
Ans. It improves communication, automates processes, enables e-commerce, and supports data management.

13. **How does ICT assist governance?**

Ans. ICT enhances transparency, service delivery, e-governance, and citizen engagement.

14. **What are ICT applications in healthcare?**

Ans. Healthcare uses ICT for patient records, telemedicine, diagnostics, and health monitoring.

15. **How is ICT used in digital media and entertainment?**

Ans. It enables content creation, streaming services, gaming, and digital marketing.

16. **What is Artificial Intelligence in ICT?**

Ans. AI enables machines to mimic human intelligence for tasks like decision-making and pattern recognition.

17. **What is the Internet of Things (IoT)?**

Ans. IoT connects physical devices to the internet, enabling smart homes, wearables, and industrial automation.

18. **How does blockchain impact communication and storage?**

Ans. Blockchain offers secure, decentralized, and transparent data management and transactions.

19. **What is quantum computing?**

Ans. Quantum computing uses quantum mechanics principles to process information faster than classical computers.

20. **What is 5G technology?**

Ans. 5G is the fifth generation of mobile networks, offering faster speeds, low latency, and improved connectivity.

21. **What are input devices?**

Ans. Devices like keyboards and mice used to enter data into a computer.

22. **What are output devices?**

Ans. Devices like monitors and printers that display or produce information from a computer.

23. **What is cloud storage?**

Ans. Storing data on remote servers accessed via the internet instead of local physical devices.

24. **What is the function of an operating system?**

Ans. It manages computer hardware and software resources and provides common services for applications.

25. **What is a network protocol?**

Ans. Rules and conventions for communication between network devices.

26. **Why is data security important in ICT?**

Ans. To protect information from unauthorized access, theft, or damage.

27. **What is e-governance?**

Ans. Using ICT to provide government services electronically to citizens and businesses.

28. **What is telemedicine?**

**Ans.** Remote diagnosis and treatment of patients using telecommunications technology.

29. **How does ICT improve business communication?**

**Ans.** Through emails, instant messaging, video conferencing, and collaboration platforms.

30. **What future trends are expected in ICT?**

**Ans.** Advancements like AI, IoT, blockchain, quantum computing, and 6G networks will shape ICT's future.



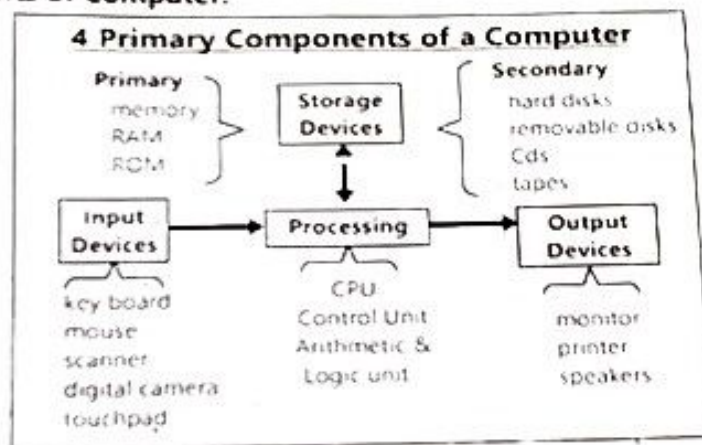
## COMPONENTS OF ICT

**Q. Define computer. Write primary components of a computer system.**

**Ans. Definition:** "A computer is an electronic device that manipulates information, or data. It has the ability to store, retrieve, and process data." OR

"Computer is an electronic machine that is used to input data, process data into useful information and store it for future use."

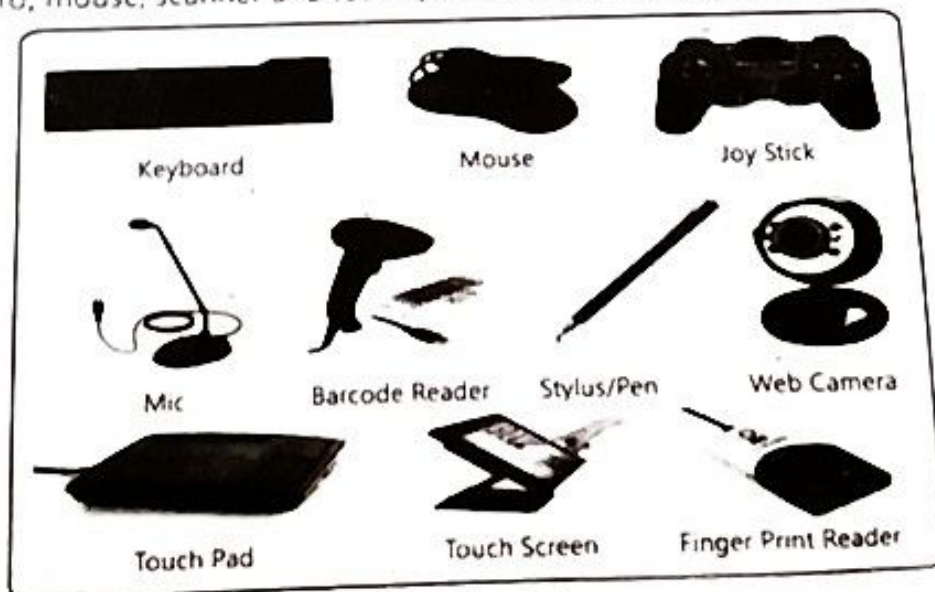
**Primary components of Computer:**



**Fig. Basic components of Computer**

### 1. Input Devices:

"An input device is a piece of hardware used to provide data to a computer used for interaction and control." It allows input of raw data to the computer for processing. For example key board, mouse, scanner and touch pad etc are commonly used input devices.



**Fig. Input devices**

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**Processing Device:**

2. "Processor is also called CPU (Central Processing Unit). It is the brain of the computer. Processor is used to process data CPU interpret and executes program instructions. All computers must have CPU."

When user input data initially data store in main memory then CPU interpret and processing. Main memory is used to store data before processing and also stores processed data until sent to the output device. Main memory is closely connected to CPU but separate from it.



Fig. CPU

**Output devices:**

3. "An **output device** is any peripheral that receives data from a computer, usually to display, projection, or physical reproduction." For example, the image shows an inkjet printer, an output device that can make a hardcopy of anything shown on the monitor. Monitors and printers are two of the most commonly used output devices used with a computer.



Fig. Output Devices

**Storage Devices:**

4. A storage device is any computing **hardware** that is used for storing, porting and extracting data files and objects. It can hold and store information both temporarily and permanently, and can be internal or external to a computer, server or any similar computing device. Commonly used storage devices are USB, Hard disk, CDs and DVDs etc.

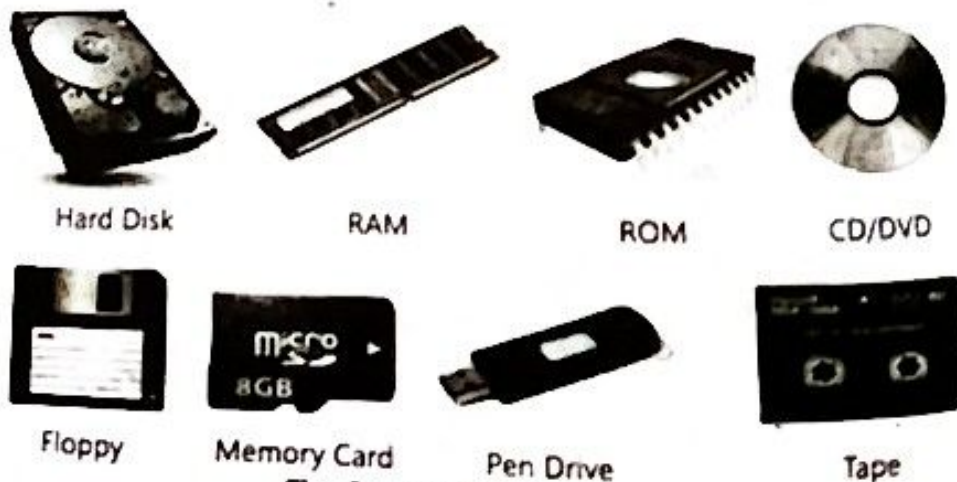
Computer storage or Memory Devices

Fig. Storage Devices

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**Q: What is software and hardware? Write their relationship.**

**Ans. Definition:**

"Software is a set of instructions, given to the computer to solve a problem".

"**Software** is a set of instructions that tell a computer what to do." Software comprises the entire set of programs, procedures, and routines associated with the operation of a computer system. A set of instructions that directs a computer's hardware to perform a task is called a program, or software program. We **cannot touch** software.

"Computer **hardware** is the collection of **physical** parts of a computer system. This includes the computer case, monitor, keyboard, and mouse." It also includes all the parts inside the computer case, such as the hard disk drive, motherboard, video card, and many others. Computer hardware is what you can physically **touch**.

#### **Relationship between Hardware and Software:**

Both are useless without each other. As like the body needs a soul to live, so does the hardware needs the software to operate. Hardware does all the work as like our body. But it's the software which instructs (Set of instructions are passed to the processor for each process to be done) the hardware what to do.

For example: when you switch on the computer it is the BIOS which first checks the hardware and booting order and then instructs the hardware to load (boot) the OS from the hard-disc or pen-drive or external drive or the cd/dvd drive. Once the instruction is passed the hardware starts to function according to the instruction received.

**Q: Difference between software and hardware.**

**Ans.**

| HARDWARE                                                                                     | SOFTWARE                                                                      |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Hardware is a physical parts computer that cause processing of data.                         | Software is a set of instructions that tells a computer what to do.           |
| It is manufactured.                                                                          | It is developed and engineered.                                               |
| Hardware cannot perform any task without software.                                           | Software cannot be executed without hardware.                                 |
| We can see and touch hardware.                                                               | We can't actually touch them.                                                 |
| It has four main categories: input device, output devices, storage, and internal components. | It is mainly divided into System software and Application software.           |
| Hardware is not affected by computer viruses.                                                | Software is affected by computer viruses.                                     |
| It cannot be transferred from one place to another electrically through network.             | It can be transferred from one place to another electrically through network. |
| If hardware is damaged, it is <b>replaced</b> with new one or we need to <b>repair</b> .     | If software is damaged, its backup copy can be <b>reinstalled</b> .           |
| Ex: Keyboard, Mouse, Monitor, Printer, CPU, Hard disk, RAM, ROM etc.                         | Ex: Ms Word, Excel, Power Point, Photoshop, MySQL etc.                        |

**Q:** Define Data and Information with example.

**Ans. Data:**

"Collection of raw facts and figures is called data".

Data is defined as facts and figures that is stored in a computer. Data is collected from user. It is the input of computer. It is unorganized.

**Information:**

"Processed data is called information." Information is an organized and processed form of data. It is used for making important decisions.

**Examples of Data and Information:**

**Example No 1:** A day's temperature, humidity, wind and speed of recorded are **Data** while percentage of weather as cold or warm is **Information**.

**Example No 2:** Students' names in a class are **Data** while names of students in alphabetic order are **Information**.

**Example No 3:** A student's subject marks are **Data** while his percentage of marks, grade and position are **Information**.

**Q:** What is data processing?

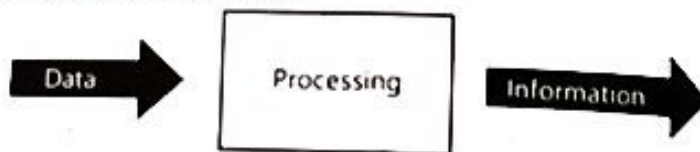
**Ans.** Process of converting data into information is called processing. Data is the input and information is the output of a processing system.

Following are the basic steps of data processing:

✓ Input

✓ Processing

✓ Output



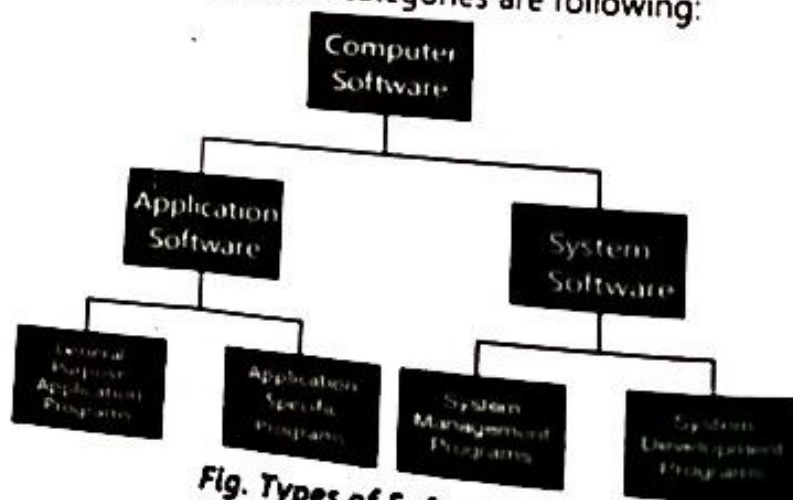
**Q:** Define software. Discuss different types of software.

**Ans. Definition:**

"Computer software, also called software, is a set of instructions and its documentations that tells a computer **what to do** or how to perform a task is called software." Computer works according to the instructions written in software.

**Types of software:**

Mainly software's are divided in two categories are following:



**Fig. Types of Software**

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## 1. System Software

"System software is a type of **computer program** that is designed to run a **computer's hardware and application programs**." If we think of the computer system as a layered model, the system software is the **interface between the hardware and user applications**. The operating system is the best-known example of system software. The OS manages all the other programs in a computer. System software coordinates the activities and functions of hardware and software, and it controls the operations of computer hardware. Some examples of system software's are

- **Windows**
- **Linux**
- **Unix**
- **Android**
- **IOS**
- **Solaris**

## 2. Application software

"An **application** is any program that is designed for the end user. Application software runs on system software. It is the software that performs a specific task for end users. **Applications software** include such things as database programs, word processors, Web browsers, games and spreadsheets."

### Categories of Application software:

Main categories of application software are follows:

**Package software:** An application program developed for sale to the general public. Packaged software is designed to appeal to a large audience of users, and although the programs may be tailored to a user's taste by setting various preferences, it can never be as individualized as custom-programmed software. For example MS office graphics data base software etc.

**Customized Software:** Custom Software is designing a software application for a specific group of customers within the organization. Such custom software is designed to address their needs precisely for a specific group of customers. For example, an application created for M.A.O College would be used only by that college and the department for which it was designed. No other college will use that software.

**Q: What is Information Technology (IT)?**

**Ans. Information technology (IT):**

"Information technology (IT) is the use of any computers, storage, networking and other physical devices to share an information".

Information Technology is the best use of computing devices to share information. For example, two computers are connected together with a wire. They form a network and the network is used to share information. This is called IT. In short, Application of computers in any field is the domain of IT.

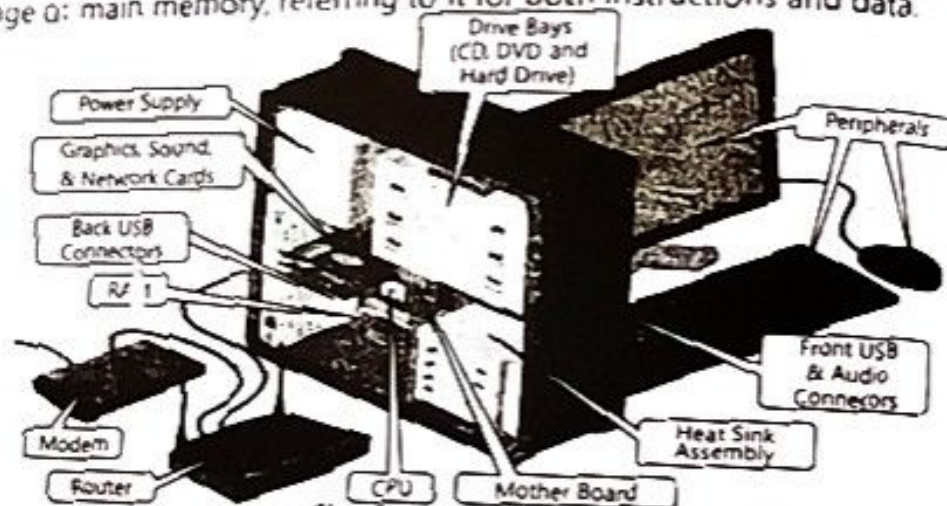
IT refers to anything related to computing technology, such as networking, hardware, software, the Internet, or the people that work with these technologies. Many companies

now have IT departments for managing the computers, networks, and other technical aspects of their businesses.

Today, IT is used in every field of life. IT is used in banks to do transactions online. It is used in Medical sciences like ECG or X-Ray etc. IT is used in office management to keep record of students or employees. IT is used in communication sector using cellular technology. Hence, the world is global village now due to internet, cellular technology, and use of IT in every field of life.

**Q: Explain CPU.**

**Ans.** CPU stands for Central Processing Unit. It is also known as microprocessor or processor. A CPU is **brain of a computer**. It is responsible for all functions and processes. The CPU contains at least one processor, which is the actual chip inside the CPU that performs calculations. For many years, most CPUs only had one processor, but now it is common for a single CPU to have at least two processors or "processing cores." CPU makes computer able to count and perform logical operations which is called processing. It processes the instructions that it collects by decoding the code in programs. Computers have two types of storage: Primary storage and secondary storage. The CPU mainly interacts with primary storage or main memory, referring to it for both instructions and data.



**Fig. System Unit**

There are four important functions of CPU,

- 1) Fetch
- 2) Decode
- 3) Execute
- 4) Write back

CPU consists of main two components known as ALU and Control Unit. The components work together to perform operations.

### 1. Control Unit:

The control unit of the Central Processing Unit regulates and integrates the operations of the computer. It selects and retrieves instructions from the main memory in proper sequence and interprets them so as to activate the other functional elements of the system at the appropriate moment.

It performs the following operations to perform:

1. Fetches instruction from main memory.
2. It interprets the instruction to find what operation is to be performed.

- 3
2. **ALU (Arithmetic Logic Unit)**  
An arithmetic unit performs operations on data of a computer.

Modern instruction takes

- Arithmetic
- Logic

**Arithmetic Unit:**

Arithmetic multiplication and

**Logic Unit:**

it performs operations that are greater than equal to

**Q: Define memory access?**

**Ans. Definition:**

"Main memory"

processor is active from secondary storage. A copy remains in the computer system.

**Structure of main memory:**

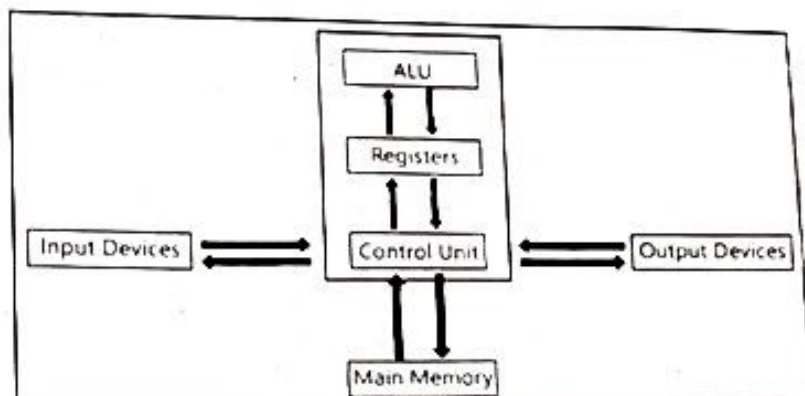
A main memory board. A main memory board stores one bit of information.

Bit cells are used to store data in bytes. The form of hexadecimal

3. It controls the execution of instructions.

## 2. **ALU (Arithmetic and Logic Unit)**

An arithmetic logic unit (**ALU**) is a digital circuit used to perform arithmetic and logic operations. It represents the fundamental building block of the central processing unit (CPU) of a computer.



Modern CPUs contain very powerful and complex ALUs. Actual execution of instruction takes place in this part. It consists of two units

- **Arithmetic Unit**
- **Logic Unit**

### **Arithmetic Unit:**

Arithmetic unit of ALU performs basic arithmetic functions like addition subtraction, multiplication and subtraction etc.

### **Logic Unit:**

It performs logical operations like comparing two data items to find which data item is greater than equal to or less than other.

**Q: Define main memory? Discuss its structure. How data stored in main memory is accessed?**

**Ans. Definition:**

"**Main memory** is the storage where programs and data are kept when the processor is actively using them." When programs and data become active, they are copied from **secondary storage** into **main memory** where the processor can interact with them. A copy remains in secondary storage. Main Memory is also known as **working area** of a computer system. It stores data and instructions **temporarily**.

### **Structure of main memory:**

A main memory is in the form of chip which is inserted into memory slot on mother board. A main memory has so many cells depending on the size of main memory. To store one bit of information, a **bit cell** is used in main memory.

Bit cells are arranged in groups. A collection of 8 bits is called Byte. And computer stores data in bytes. Each Byte in the memory has a unique address. The address is in the form of hexadecimal number.

Q: Define RAM? Describe different types of RAM.

Ans. Definition:

RAM (Random Access Memory) is a part of computer's Main Memory which is directly accessible by CPU. RAM is used to **Read and Write** data into it which is accessed by CPU randomly. RAM is **volatile** in nature, it means if the power goes off, the stored information is lost. RAM is used to store the data that is **currently processed** by the CPU. Every data or instruction are loaded on RAM for execution. It stores data and instructions temporarily.

Integrated RAM chips are available in two form:

1. SRAM(Static RAM)
2. DRAM(Dynamic RAM)

The block diagram of RAM Types is given below.

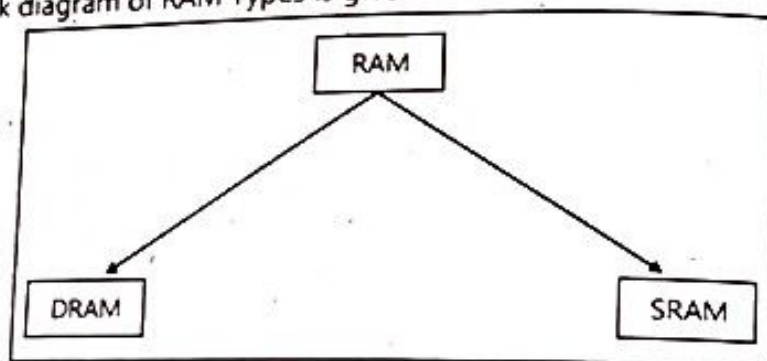


Fig: Types of RAM

### 1. DRAM (Dynamic Random Access Memory):

DRAM stands for Dynamic Random Access Memory. The term dynamic indicates that the memory must be **constantly refreshed** or it will lose its content. DRAM takes time to refresh. Moreover, DRAM is an off-chip memory which has a **large access time**. Therefore DRAM is **slower** than SRAM. DRAM is available in **larger storage capacity** compared to SRAM. DRAM is **cheap** and economic. DRAM requires **reduced power** consumption as the information stored in the capacitor.

### 2. SRAM (Static Random Access Memory):

SRAM stands for Static Random Access Memory. SRAM is **faster** than DRAM because it is on-chip memory device and takes **less access time**. It is costly than DRAM. SRAM is said to be static as it does not need to be refreshed. It comes with **less storage capacity**. SRAM needs a constant power supply, which means it consumes **more power**.

However, both types of RAM are **volatile**, meaning that they lose their content when the power is turned off.



Q: List out di  
Ans.

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Q: Define R

Ans. Definitio

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Types of ROM

RAM ha

1. PROM:

PROM  
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program once  
change it again

2. EPROM

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times. Program  
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3. EEPROM

It stand  
erase and rew  
types of ROM

Fig. SRAM

Q: List out difference between SRAM and DRAM:

Ans.

| SRAM                                                                            | DRAM                                                                                       |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| SRAM has lower access time, which is <b>faster</b> compared to DRAM.            | DRAM has a higher access time. It is <b>slower</b> than SRAM.                              |
| SRAM is <b>costlier</b> than DRAM.                                              | DRAM <b>cost is lesser</b> compared to SRAM.                                               |
| SRAM needs a constant power supply, which means it consumes <b>more power</b> . | DRAM requires <b>reduced power</b> consumption as the information stored in the capacitor. |
| SRAM offers low packaging density.                                              | DRAM offers a high packaging density.                                                      |
| The storage capacity of SRAM is 1MB to 16MB which is <b>less</b> than DRAM.     | The storage capacity of DRAM is 1 GB to 16GB which is <b>greater</b> than SRAM.            |

Q: Define ROM? Discuss its different types.

Ans. **Definition:**

ROM (Read Only Memory) is a memory device or storage medium that stores information **permanently**. It is also the primary memory unit of a computer along with the random access memory (RAM). It is called **read only memory** as we can only read the programs and data stored on it but cannot write on it. It is restricted to reading words that are permanently stored within the unit. It is **non-volatile** memory as data is stored even if power (electricity) goes off.

The manufacturer of ROM fills the programs into the ROM **at the time of manufacturing** the ROM. After this, the content of the ROM cannot be altered, which means you cannot reprogram, rewrite, or erase its content later. However, there are some types of ROM where you can modify the data.

### Types of ROM

RAM has three main types based on functionality. These are as follow:

#### 1. PROM:

PROM stands for **Programmable Read only Memory**. Programmable ROM is essentially a blank version of ROM that you can purchase and software engineer can **program once** with the help of a **special tool**. Once you have programmed, then you cannot change it again.

#### 2. EPROM:

**Erasable Programmable ROM** chips allow you to **write and rewrite** them many times. Programmer can erase the data with **optical special device**. So if you want to erase the data, you must purchase the optical special device.

#### 3. EEPROM:

It stands for **Electrically Erasable Programmable ROM**. It provides the facility to erase and rewrite the data using electric field. EEPROMs have several advantages over other types of ROM. Unlike the earlier form, you can rewrite EEPROM without dedicated

equipment and without removing it from the hardware. You do not have to erase and rewire everything to make a single edit.

Q: List out difference between RAM and ROM:

| RAM                                                                                 | ROM                                                                                    |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| RAM stands for <b>Random Access Memory</b>                                          | ROM stands for <b>Read Only Memory</b>                                                 |
| RAM data is <b>volatile</b> . Data is present till power supply is present.         | ROM data is <b>non-volatile</b> . Data remains even after power supply is not present. |
| RAM data can be <b>read, erased or modified</b> .                                   | ROM data is <b>read only</b> .                                                         |
| RAM is used to store data that CPU needs for <b>current instruction</b> processing. | ROM is used to store data that is needed to <b>bootstrap</b> the computer.             |
| RAM speed is quite <b>high</b> .                                                    | ROM speed is <b>slower</b> than RAM.                                                   |
| CPU can access data stored on RAM.                                                  | Data to be copied from ROM to RAM so that CPU can access its data.                     |
| RAM memory size is <b>large</b> .                                                   | ROM memory size is generally <b>small</b> .                                            |

Q: Define Cache memory. How does it affect the working of CPU?

Ans. Cache is pronounced as "Cash". Cache is a **small** in capacity and very **fast** memory which is part of the CPU which is physically closer to the CPU than RAM is. It is installed on motherboard close to processor. It stores the data which user access repeatedly. The more cache there is, the more data can be stored closer to the CPU.

Cache memory is beneficial because:

- Cache memory holds **frequently used instructions/data** which the processor may require next and it is faster access memory than RAM, since it is on the same chip as the processor.
- This **reduces the time** for data/ information retrieval as CU can access data from cache within no time.

If the CPU has more cache, it takes less time to access instructions and as a result programs may run faster.

Q: Define input? Write use of input devices and list out different input devices.

Ans. Definition:

"Any information or data sent to a computer for processing is considered input".  
 "Anything given to computer is called input."

Input Devices:

"An input device is a piece of hardware used to provide data to a computer used for interaction and control. It allows input of raw data to the computer for processing." Basically input devices take data from user and convert it in a form that understandable by computer. There are three general types of input devices.

Keyb  
Poin  
Data

Q: Define key

Ans. Definition:

"A compu  
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**QWERTY** keyboa  
standard keyboa

Row K

Row E

Row D

Row C

Row B

Row A

Following are the

- 1.
- 2.
- 3.
- 4.

1. **Function**  
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2. **Main**  
keypad. It co  
software that

Some  
keys and cor

a) **ESC**  
keyb

b) **Tab**  
keyb

equipment and without removing it from the hardware. You do not have to erase and rewrite everything to make a single edit.

**Q: List out difference between RAM and ROM:**

**Ans.**

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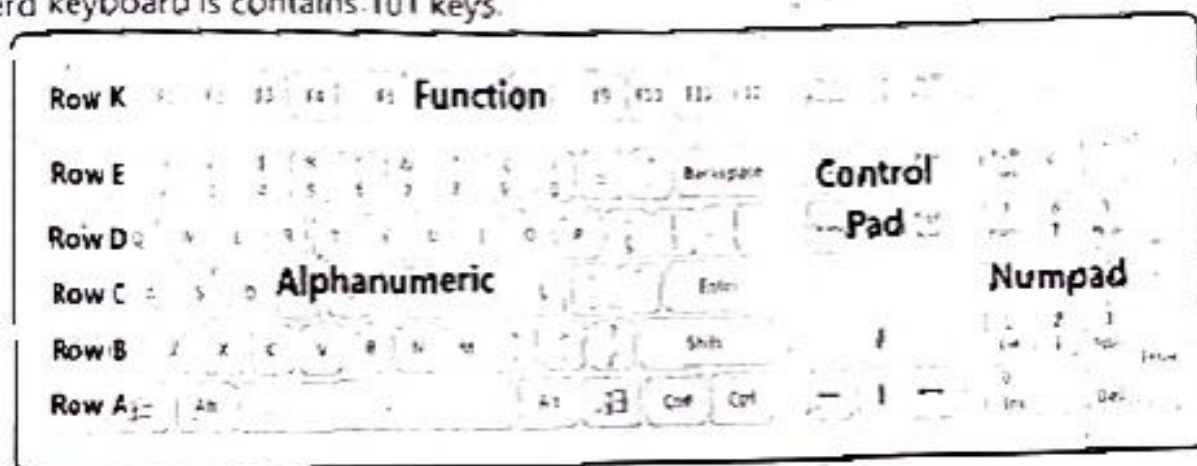
"An input device is a piece of hardware used to provide data to a computer used for interaction and control. It allows input of raw data to the computer for processing." Basically input devices take data from user and convert it in a form that understandable by computer. There are three general types of input devices.

- Keyboard
- Pointing devices
- Data entry Devices

**Q: Define keyboard? Discuss Enhanced keyboard and its segments.**

**Ans. Definition:**

"A **computer keyboard** is an input device used to enter characters and functions into the computer system by pressing buttons, or keys. It is the primary device used to enter text." A keyboard typically contains keys for individual letters, numbers and special characters, as well as keys for specific functions. Generally used standard keyboard is called **QWERTY** keyboard because first six alphabets on the top row is Q, W, E, R, T, and Y, a standard keyboard is contains 101 keys.



**Fig. Keyboard**

Following are the main parts of keyboard:

1. Function keys (In Top)
2. Main keyboard (In Center)
3. Numeric keys (Right)
4. Additional keys

1. **Function Keys:** The function keys on a computer keyboard labelled F1 through F12, are keys that have a special function defined by a currently running program or by the operating system. They can be combined with the **Ctrl** or **Alt** keys.

2. **Main Keyboard:** Main keyboard includes the keys found on an alphanumeric keypad. It contains some special keys. Special keys have different uses that depend on software that used.

Some keyboards have some additional in the center of **alphanumeric and numeric keys** and contain status light in upper right corner called **101-enhanced keyboard**.

- a) **ESC Key:** Short for Escape, Esc is a key found on the top-left corner of a computer keyboard and allows the user to abort, cancel, or close an operation.
- b) **Tab Key:** The tab key "Tab" ? (abbreviation of tabulator key or tabular key) on a keyboard is used to advance the cursor to the next tab stop.

22

- c) **Caps Lock Key:** Caps Lock is a toggle key on a computer keyboard that causes all letters typed to be in uppercase until disabled. For example, when Caps Lock is enabled, the typed letters appear "LIKE THIS." When disabled, they appear "like this."
- d) **Shift key:** The Shift key is a keyboard modifier key that allows a user to type a single capital letter and change the top number keys to a symbol. For example, pressing and holding the Shift key while pressing "a" generate a capital "A" and pressing the Shift and the number "1" creates an exclamation mark on US keyboards.
- e) **Control Key:** a Control key "Ctrl" is a modifier key which, when pressed in conjunction with another key, performs a special operation (for example, Ctrl + C); similar to the Shift key, the Control key rarely performs any function when pressed by itself.
- f) **Alt Key:** The "Alt" key on a computer keyboard is used to change (alternate) the function of other pressed keys. Thus, the Alt key is a modifier key, used in a similar fashion to the "Shift" key.
- g) **Backspace Key:** Backspace is the keyboard key that originally pushed one position backwards and in modern computer systems moves the display cursor one position backwards, deletes the character at left position, and shifts back the text after that position by one position.
- h) **Del Key:** It is used to delete one character from right side in the text editor. Moreover, it can delete all the selected text in the text editor.
- i) **Enter Key:** It is alternatively known as a Return key. The Enter key sends the cursor to the next line or executes a command or operation. Most full-sized PC keyboards have two Enter keys; one above the right Shift key and another on the bottom right of the numeric keypad.
- j) **Numeric Keys:** A separate set of keys on some keyboards that contain the numbers 0 through 9 and a decimal point arranged as on an adding machine. Numeric keypads make it easier to enter large amounts of numeric data. It is used for two purposes depending on the status of the NumLock key. If NumLock is "ON" the keys are used to enter numeric data and arithmetic operations and if it is "OFF" then the keys are used to perform cursor control functions.

Q: **Define pointing Devices? List out different pointing devices.** 10-09-23

Ans. **Definition:**

"Pointing device is an input device that allows user to control and provide data to the computer using physical gestures by moving a hand-held mouse or similar devices." The use of pointing devices is to control the position of the cursor or pointer on the screen and allow the user to select options displayed on the screen. Commonly pointing devices are mouse, and it includes the pointing stick, touch pad, digitizer,

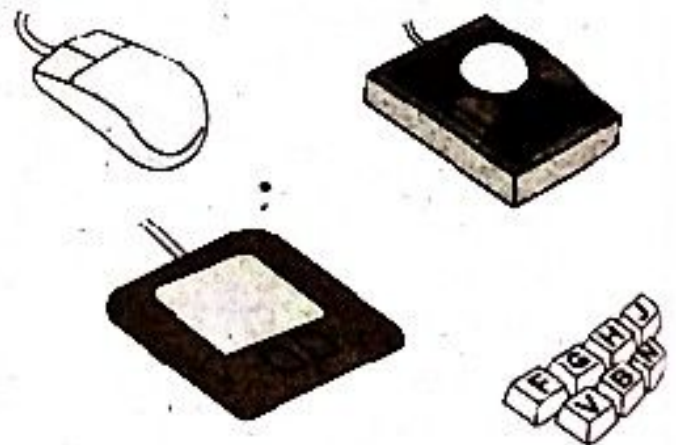
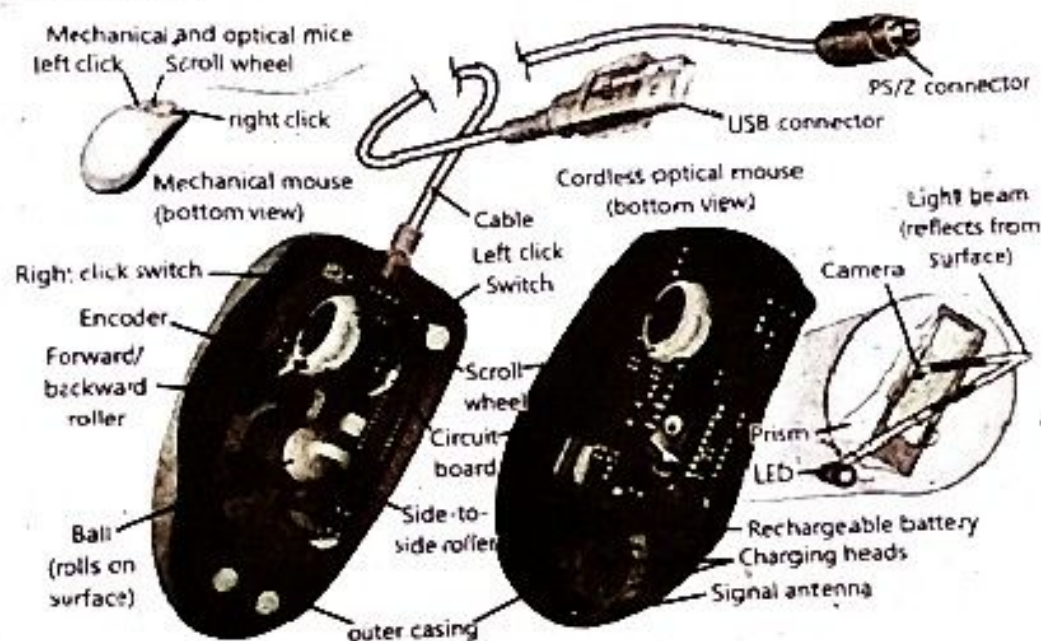


Fig. Pointing Devices

touch screen, light pen, joystick, pen based system, and others.

✓ **Mouse:** "A **mouse** is a handheld hardware input device that controls a cursor on screen and can move and select text, icons, files, and folders." For desktop computers, the **mouse** is placed on a flat surface such as a **mouse pad** or a desk and is placed in front of your computer. Mouse consist of two type which is mechanical mouse and optical mouse. The **mechanical mouse** has a ball inside the mouse and it touches a surface and rolls with the mouse. The **optical mouse** use laser beams and special chips to encode data for the computer.

Every mouse has two button on it; Left Button and Right Button. When we click on left button, it is called **left click**. When we click on right button, it is called **right click**. It has scroll wheel as well in between the two buttons which is used to scroll up and down the document pages or lines.



**Fig. Mouse**

**Trackball:** A trackball is a pointing device which has a ball on upper side. We move the ball with the finger. The movement of the ball sends the information to computer. The user rolls the ball with the thumb, fingers, or the palm of the hand to move a cursor on screen. The major difference between trackball and mouse is that ball is used on upper side of trackball whereas it is used in downside of mouse.

**There are several advantages a trackball has over a mouse.**

- A trackball can be used whether you're left handed or right handed.
- Trackball does not need to move across the desk.
- Trackball can be used in a smaller space.
- Trackball is more accurate.
- Trackball is easy to use.



Fig. Trackball

**Joystick:** "A joystick is an input device that can be used for controlling the movement of the cursor or a pointer in a computer device. The pointer/cursor movement is controlled by maneuvering a lever (handle) on the joystick." The input device is mostly used for gaming applications and, sometimes, in graphics applications. A joystick also can be helpful as an input device for people with movement disabilities.

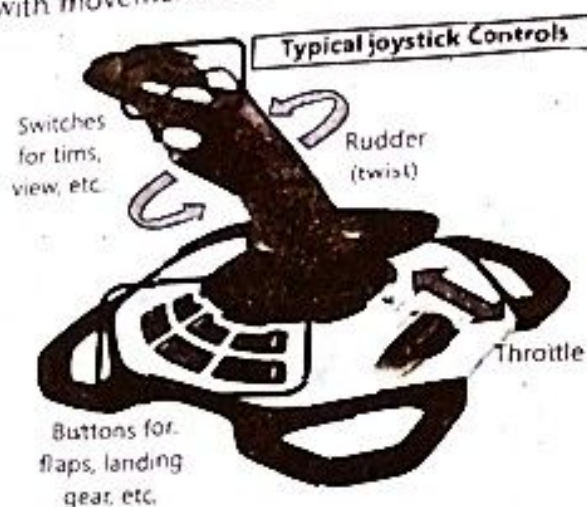


Fig. Joystick

**Touchpad:** A touchpad (also called a track pad) is a type of input device for computers that does the same things as a computer mouse. It is made up of a flat, touch-sensitive surface which the user slides one or more fingers on to move the cursor on the screen. Next to the touchpad are push-buttons that work just like mouse buttons, including left-clicking and right-clicking. In some newer touchpads, there are no actual buttons, and clicking is done by pushing near the button of the touchpad itself.



Fig. Touchpad

**Pointing Stick:** Pointing stick is a pressure-sensitive small nub used like a joystick. It is located between the keys on a laptop keyboard that allows the user to control the pointer by directing the stick with one finger. Commonly found in between 'G' and 'H' key, and also at below the spacebar.



Fig. Pointing Stick

**Touch screen:** Touch screen is a device embedded into the screen of the LCD monitor screen. It use with an interaction between the screen by physically touch of finger. It very sensitized to detect touch on the screen. Generally used in ATMs, voting machines, Smartphone, tables, laptops, and e-books.



Fig. Touch Screen

**Multi touch Screens:** Multi touch screen is a device that allow two or more fingers or other gestures such a pinching motions to be recognized as input at any one time. It allows pinching and stretching gestures on the screen to control zooming.



Fig. Multi touch screen

**Pen based Input:** It is user-interface by pen-like device for input. The device is called stylus. All the selection and work is done with stylus (pen like device). It commonly use in smart phones, wireless tables personal computer, PDAs, and others.



Fig. Pen Based Input

**Light Pen:** A light-sensitive pen like device that uses a wired connection to a computer terminal to transmit data. It uses light. The concept is like a mouse, just bring the pen to desired location and press a button to identify the screen location. It most popular among graphics artists, engineers, and in situations that require covered hands.



Fig. Light Pen

**Digitizer:** Digitizer is an electronic pen or puck that used to convert drawings and photos to digital data that can be process by computer. It often found in special tables use for design and drawing.



Fig. Digitizer

**Digital Pen:** Digital pen is an electronic device which consists of microchip in the pen to capture the handwriting or brush strokes of user and then convert it into **digital ink** that can be written on a paper. The writing then sent to computer as image file.

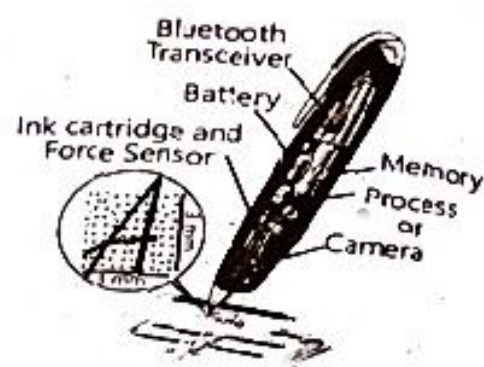


Fig. Digital Pen

✓ 11-09-25  
Q. Define source data entry devices? Enlist different categories of source data entry devices.

Ans. Definition: "The devices that are used for direct data entry to the computer system are called *source data-entry devices*. The devices are also known as *direct input devices*." These devices provide quick and efficient way to input data into the computer. Following is a list of commonly used source data entry devices:

#### Scanning Devices

1. Barcode Reader
2. Fax machine
3. Image scanner
4. Mark and character recognition devices such as;
  - MICR (Magnetic-Ink Character Recognition)
  - OMR (Optical Mark Recognition)
  - OCR (Optical Character Recognition)
  - Magnetic-Strip Card
  - Smart Card

### 1- Barcode Reader

Barcode is a unique identification code. It consists of light and dark vertical lines with spaces between them of different widths. Barcode is printed on most of the manufactured products. It represents the information about the product. Actually, the name of product, its price and other descriptions are coded as barcode. There are several bar-coding schemes or systems in use. The most popular and commonly used barcode is UPC (Universal Product Code).



Fig. Barcode Reader

Barcode reader is a special scanning device. It is used for reading (or decoding) barcode printed on the product or book. It uses LASER beam to scan or read the barcode. It reads the barcode and translates it into digital signals. These signals are input to the computer for further processing.

### 2- Fax machine

"A fax machine is also known as facsimile transmission machine. It is used to send and receive information through telephone line."

The fax machine at sending side scans information printed on paper. It converts the



Fig. Fax Machine

The fax machine at sending side scans information printed on paper. It converts the information into digital/analog signals and sends these signals over telephone line to a fax machine at receiving side. The fax machine at receiving side receives information in the form of digital/analog signals. It converts these signals into original form of the information and print on the paper.

### 3- Image Scanner

"Image scanner is also called graphic scanner. It reads text, drawing and photographs printed on the paper. It converts them into digital form and stores inside the computer for further processing."



Fig. Image Scanner

Image scanner scans the image of information printed on paper with light. It breaks image into light and dark dots if image is black and white. In case of color image, it breaks the image into color dots. These dots are then converted into digital form. In this way, the image is stored into the computer. The technique of representing a graphic image as a matrix of dots is called *raster graphics*.

4- Mark and Character Recognition Devices **MCR :**

"Mark and character recognition devices are input devices. They are used to scan (or read) information printed on the paper." These devices read the information and convert into digital signals. These signals are input to the computer for further processing. The commonly used mark and character recognition devices are described below.

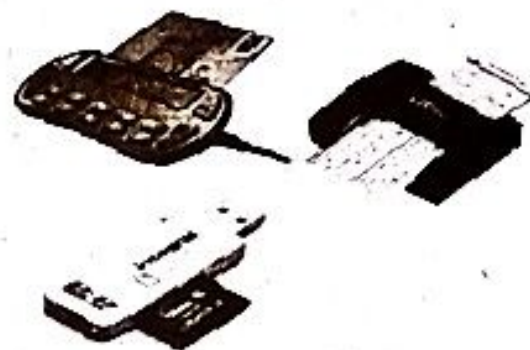


Fig. Mark and Character Recognition Devices

(i) **MICR** - *magnetic ink character Recognition*.

"MICR stands for Magnetic-Ink Character Recognition. It is a special scanner. It reads the character printed with **magnetic ink** and converts them into digital signals. These signals are then input to the computer for further processing."

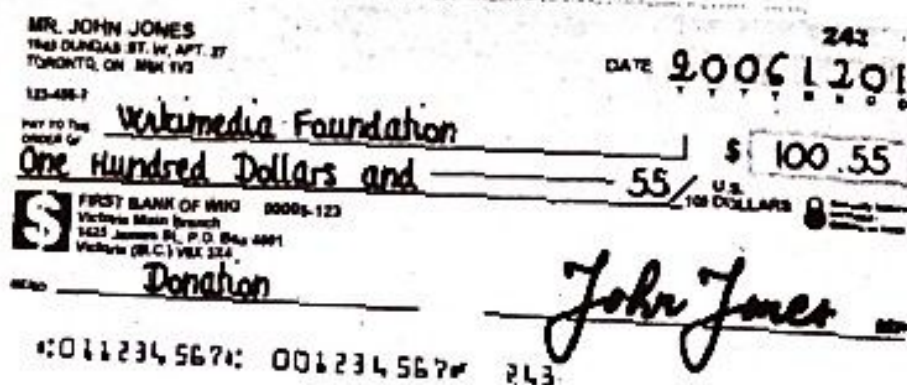


Fig. MICR

MICR devices are normally used in banks for processing check. On a check, the bank code, customer account number, and check number is printed with magnetic ink. The customer check is given to the MICR device. It reads the information printed on the check and sends to the computer for processing. It is magnetic ink which authenticates the check.

(ii) **OMR** :

"OMR stands for Optical Marks Recognition. OMR device is used to read handwritten marks or symbols printed on the paper." It uses a light beam to scan the marks and convert them into digital signals. These signals are then input to the computer for further processing.



Fig. OMR

OMR devices is usually used in education field to check (or mark) the objective type answers of the questions. For example, it is used to check and process the test marks of various exams. The pre-printed answer sheet is given to the OMR device. It scans the answer sheet and sends the information to the computer. The computer automatically evaluates and calculates the result according to the program instructions.

### (iii) OCR

"OCR stands for Optical Character Recognition. OCR device reads preprinted characters in a particular font and converts them into digital code." The characters printed on paper may be typewritten or handwritten. In case of handwritten, the characters must be of standard size. But in case of typewritten, the characters must have the special type font called an OCR font. The accepted OCR fonts are OCR-A (American standard) and OCR-B (European standard).

The OCR devices are commonly used to process utility bills and price code printed on the products in departmental stores.

### (iv) Magnetic-Strip Card

A magnetic-stripe card has a strip of magnetically encoded information on its back. Usually, the magnetic strip is used in ATM Cards. This card is used for payment or transactions at various departmental stores. This magnetic strip card is swapped into a card reader for reading information encoded on the strip. It includes all the necessary information needed for a transaction.



Fig. OCR



Fig. Magnetic-Strip Card

### (v) Smart Card

Smart card looks like a credit card or ATM card. This card contains a thin microprocessor and memory chip that is embedded in it. The card is inserted into a card reader. The card reader reads information stored on the card. It can also store and update some basic information. A Smart ID card and an ATM card are examples of smart card. Smart ID cards holds all the necessary information.



Fig. Smart Card

### Audio/Video Devices

1. Audio input devices
2. Video input devices
3. Digital Camera

### 1- Audio Input Device

"Audio input is the process of entering any sounds into the computer. It may include speech, and music etc. The device that is used to enter/record sound (audio input) into computer is called *audio input device*."



Fig. Audio Input Device

Microphone is most commonly used as an audio input device. The main purpose of audio input device is to provide digital input for multimedia computers.

Audio input is in the form of analog signals. It must be converted into digital before storing it into the computer. For this purpose, there is a special circuit board in the computer. This circuit board is known as *audio board*. Audio input device is connected to the audio board. The audio board converts the analog signals of sound into digital form and stores it for further processing. The audio board can also convert digital signals into analog signals.

### 2 Video Input Device

"The process of entering video data into a computer is called *video input*. Video input consists of motion pictures or movie films. The devices that are used to enter video input into computer are called *video input devices*." Mostly these devices are used for recording movie films and video images. VCR and video camera are examples of video input devices.



Fig. Video Input Devices

Video input is in the form of analog signals. It must be converted into digital before storing it into the computer. For this purpose, there is a special circuit board in the computer. This circuit board is known as *video-capture card*. Video input device is connected to the video capture card. This card converts the video signals into digital signals that a computer can use.

### 3 Digital Camera

A digital camera is an input device. It takes pictures and stores them in digital form. It does not use traditional camera film. It uses a light-sensitive processor chip to capture photographic images in digital form.



Fig. Digital Camera

These images are stored on a small diskette inserted in the camera or on flash memory chip. The stored pictures or photographic images can then be loaded in the computer from the disk (or flash memory chip) for further processing.

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Ans.

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Differ

**Q:** Define output? What are different types of output? Enlist different types of output devices.

**Ans. Definition:** "Information generated by a computer is referred to as output. This includes the information produced at a software level, such as the result of a calculation, or at a physical level, such as a printed document". The most commonly used output device is the computer's monitor, which displays data on a screen.

Two types of output are there:

**Soft copy:** Soft copy is an electronic display of digital information, such as files viewed on the pc monitor. A digital picture file, a digital Word document, an email attachment are all examples of **soft copy**.

**Hard Copy:** Once soft copy is printed, they are called hard copy.

**Output Devices:** An output device is any piece of computer hardware equipment which converts information into human-readable form. It can be text, graphics, audio, and video. Examples of output devices are Monitors, printers, plotters, and Speakers etc.

The major difference between output device and input device is that output device is used to produce information produced by computer whereas the input device is used to capture input data.

**Q:** Define display screens? Which features distinguish one display screen to another? *Monitors/*

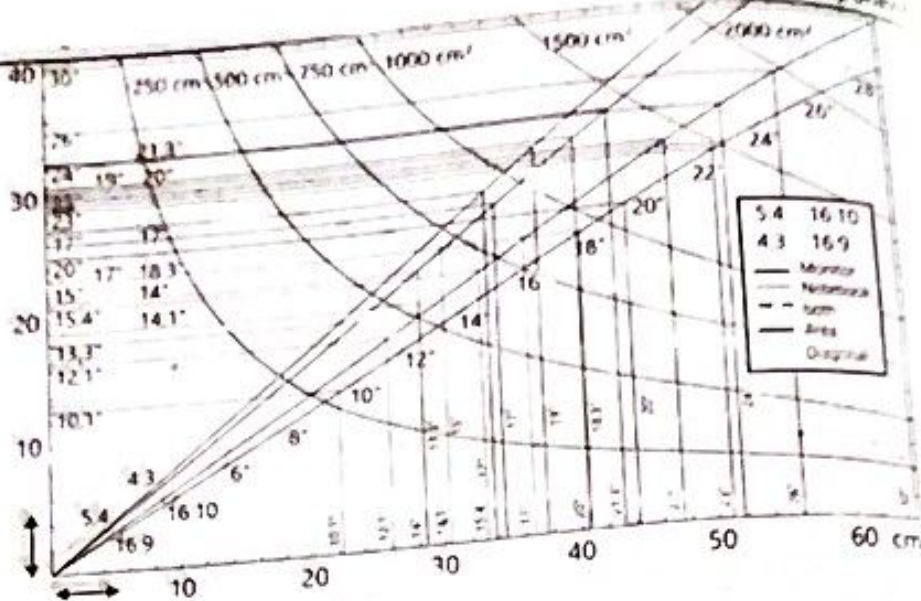
**Ans. Definition:** "Display screen is an output device that displays text, video and graphics. It is also called monitor or screen. Display screens are most common output device." Different display screens are available some have touch screen.



Different display screens can be distinguished on the basis of the following features.

**Size:**

The size of a screen is usually described by the length of its diagonal, which is the distance between opposite corners, usually in inches. Desktop monitors are available in range from 17 to 30 inches large one also available and laptop screens are available are usually 14 to 17 inches.



### Resolution:

The number of horizontal and vertical pixels on a display screen. The more pixels, the more information is visible without scrolling. Screen resolutions have a pixel count of 1600x1200, which means 1,600 horizontal pixels and 1,200 vertical pixels.

### Color:

A display monitor is capable of displaying many colors. In contrast, a monochrome monitor can display only two colors -- one for the background and one for the foreground. Color monitors implement the RGB color model by using three different phosphors that appear red, green, and blue when activated. All the colors are formed by mixing these three colors.

**Q:** Explain different types of display screens.

**Ans.** Display screen are of two types are as follows

1. **CRT Monitor** → *display video*

The Cathode-Ray Tube (CRT) is a vacuum tube that contains one or more electron guns and a phosphorescent screen, and is used to display images. It modulates, accelerates, and deflects electron beam(s) onto the screen to create the images.

### Working:

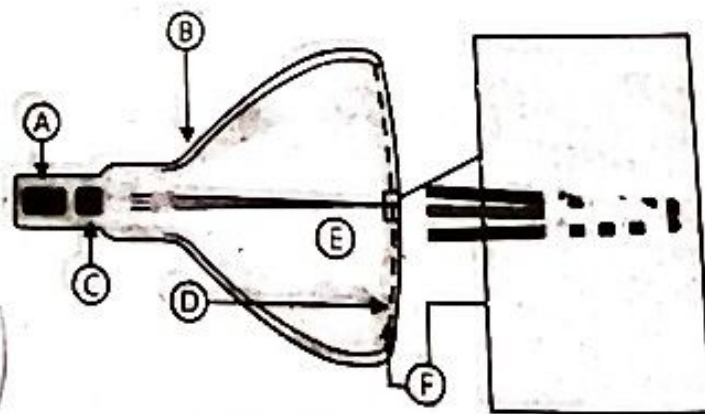


Fig. CRT

*Flat panel display*

- Light weight
- portable
- less harmful
- Expensive

*CRT*

- weight
- not portable
- More harmful
- In-expensive

A CRT monitor that glow when struck by an image in a cathode ray vacuum created by an electron gun that is negative. The anode screen is coated with a beam.

### 2. Flat Panel

A thin alternative to the electroluminescent panel display is a type of Flat panel.

### I. Liquid

Liquid crystal material substrate (diagonal) to allow colors and images.

### II. Gas

It uses gas between neon bulbs.

### Advantage

**Q:** D

**Ans.** A

document

Printer

Classification

A CRT monitor contains millions of tiny **red, green, and blue** (RGB) phosphor dots that glow when struck by an electron beam that travels across the screen to create a visible image. In a cathode ray tube, the "**cathode**" is a heated filament. The heated filament is in a vacuum created inside a glass "tube." The "ray" is a stream of electrons generated by an electron gun that naturally pour off a heated cathode into the vacuum. Electrons are negative. The anode is positive, so it attracts the electrons pouring off the cathode. This screen is coated with **phosphor**, an organic material that glows when struck by the electron beam.

## 2. Flat Panel Display: *→ multiple devices*

A thin and lightweight video display used in computer monitors and televisions as an alternative to the cathode-ray tube. Flat-panel displays often employ **liquid crystals** or electroluminescent materials such as light-emitting diodes. It is also called **flat screen**. Flat panel display is made up of two plates of glass. Plates contain a substance. There are two types of Flat panel display.

### I. Liquid Crystal Display (LCD):

Liquid crystal display is made of two pieces of **polarized glass** that contain a **liquid crystal** material between them. A backlight creates light that passes through the first substrate (glass). At the same time, electrical currents cause the liquid crystal molecules to align to allow varying levels of light to pass through to the second glass and create the colors and images that you see.

### II. Gas Plasma Display:

It uses gas between two glasses. A **gas-plasma display** works by sandwiching neon gas between two plates. You can think of a **gas-plasma display** as a collection of very small neon bulbs. It is older technology and now it is out of market.

### Advantages of Flat Panel Display:

- It requires less power (electricity).
- Its size is smaller.
- It has low weight than CRT.
- It produces high quality images.
- It has lower side effect on eye sight.

**Q: Define Printer? Discuss different classification of Printers.** *Imp*

**Ans.** A **printer** is an output **device** that prints paper documents. This includes text documents, images, or a combination of both. Output of the printer is called **hard copy**. **Printer resolution** is commonly measured in DPI (Dot Per Inch).

### Classification of Printers:

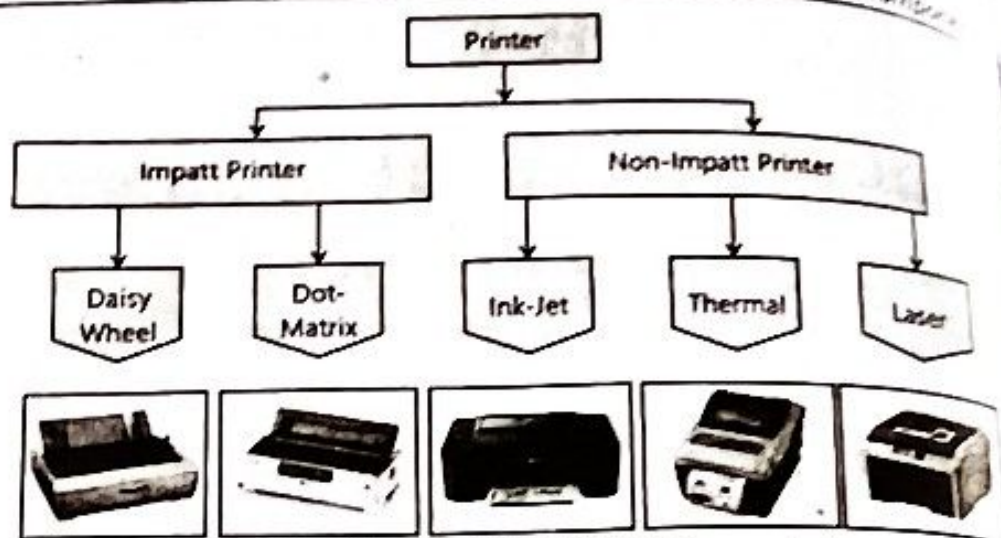


Fig. Classification of Printers

1. **Impact Printer:** *low quality → low speed/price/performance*

**Impact printer** works by creating a direct contact between ink ribbon and paper. This type of printer is **noisy** yet popular. **Impact printer** has mechanical moving parts and conduct printing. Examples: Dot-matrix printer, Daisy-wheel printer, and line printer.

#### Types of Impact Printer

- **Dot Matrix Printer**
- **Daisy Wheel Printer**
- **Line Printer** *character by character*

**Dot Matrix Printer:** A dot matrix printer is an impact printer that prints using a number of **pins** or wires. Typically the pins or wires are arranged in one or several **Columns and Rows**. The pins strike an ink-coated ribbon and force contact between ribbon and the paper, so that each pin makes a small dot on the paper. The combination of these dots forms a dot matrix image.

It writes **one character at a time**. The quality of dot matrix printers depends on the vertical and horizontal resolution and the ability of the printer to overlap adjacent dots. 9-pin and 24-pin are common; this specifies the number of pins in a specific vertically aligned space. With 24-pin printers, the horizontal movement can slightly overlap dots, producing visually superior quality. More pins will create more qualitative print.

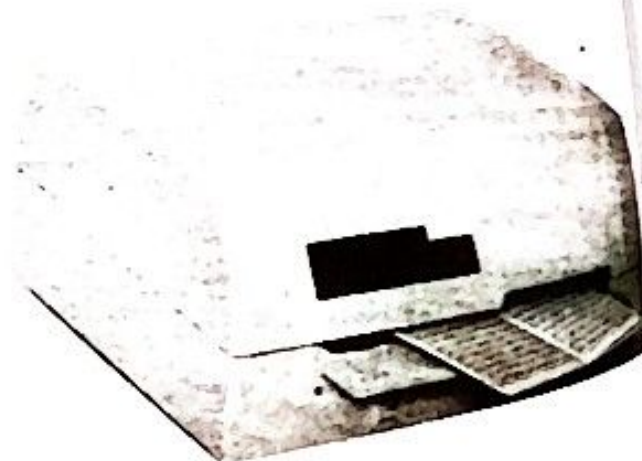


Fig. Dot Matrix Printer

ILMI Applications

#### Daisy Wheel Printer

A daisy wheel printer, like a dot matrix printer, uses an ink ribbon. A daisy wheel printer rotates the disk ribbon to create characters. The speed is slow.

Daisy wheel printers are used because of their low cost. They have less noise than impact printers.

#### Line Printer

A line printer prints an entire line of text at once. It is an early line printer.

Line printers are used for digital data processing (approx. 100 to 2500 lines per minute).

often used for batch processing but even today they are used.

Q:

Ans. A line printer is a mechanical printer that strikes the paper to create a line of text.

Type

**Daisy Wheel Printer:** — character by character

A daisy wheel printer is an early type of impact printer invented in 1969 by David S. Lee at Diablo Data Systems. The printer uses a metal or plastic disk containing each of the letters, numbers, and other characters at the end of each petal. While printing, the printer rotates the disk to each character and then using a hammer strikes each character into an ink ribbon to create the character on paper. It also prints a single character a time. Hence the speed is slow.

Daisy wheel printers were popular in the 1970s and 1980s, but are no longer used because of they have a low quality of print, are very slow and noisy than non-impact printers. They have less cost.



Fig. Daisy Wheel Printer

**Line Printer:**

A line printer prints one entire line of text before advancing to another line. Most early line printers were impact printers.

Line printers are mostly associated with unit record equipment and the early days of digital computing, but the technology is still in use. Print speeds of 600 lines per minute (approximately 10 pages per minute) were achieved in the 1950s, later increasing to as much as 1200 lpm. Line printers print a complete line at a time and have speed in the range of 150 to 2500 lines per minute.

As the line printer prints each line at a time, so the speed is high. Line printers are often used for heavy duty. They are also economic. Although LASER printers are used mostly, but even these line printers are used in some businesses due to low cost.

**Q:** Define non-impact printer? Discuss different types of non-impact printer.

Ans.  
Working mechanism

**Definition:** A non-impact printer prints characters and graphics on a piece of paper without striking the paper. Some of these printers use spray ink while others use heat and pressure to create images. These printers are faster than impact printer.

**Types of non-impact Printers**

Different types of non-impact printers are as follows:

1. Laser Printer
2. Inkjet Printer
3. Thermal Printer

high quality (but different) → Expensive → Fast speed

### 1. Laser Printer:

Laser stands for Light Amplification by Stimulated Emission of Radiation. The laser printer is a non-impact printer. Its working is similar to photocopying machine. It uses laser beams to burn special powder on page to create a permanent image. The power is contained in toner.

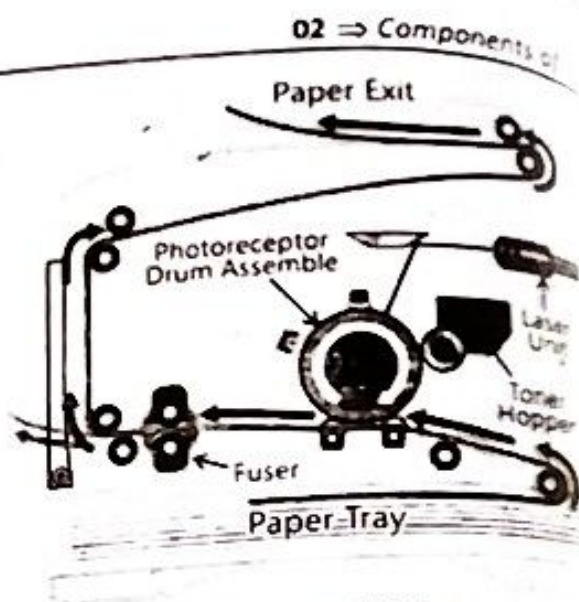


Fig. Laser Printer

Laser printer prints complete **page at a time**. It is also known as page printer. The printing speed of laser printers is 5 to 300 pages per minute (ppm). The speed depends on the contents being printed. The plain text is printed faster than graphics. Laser printer prints text and graphics in a high quality resolution, a typical laser printer provides the resolution of **1200dpi** or more. Laser printer prints at higher speed than inkjet and dot matrix printers. It does not make much noise. Laser printer is mostly used in business field.

### 2. Inkjet Printer:

"An ink-jet printer is a type of non-impact printer. It prints characters and graphics by **spraying tiny drops of liquid ink** on paper. These printers can produce quality text and graphics in both black-and-white and color including photos." A typical inkjet printer provides resolution of 300 dots per inch. The latest inkjet printers provide higher resolution.

The price of inkjet printers is less than laser printers. However, they are slower than laser printers. Most inkjet printers can print from 10 to 35 pages per minute. However, they print graphics and colors at slower rate. The print quality of inkjet printers is higher than dot matrix printer but slower than Laser Printer.

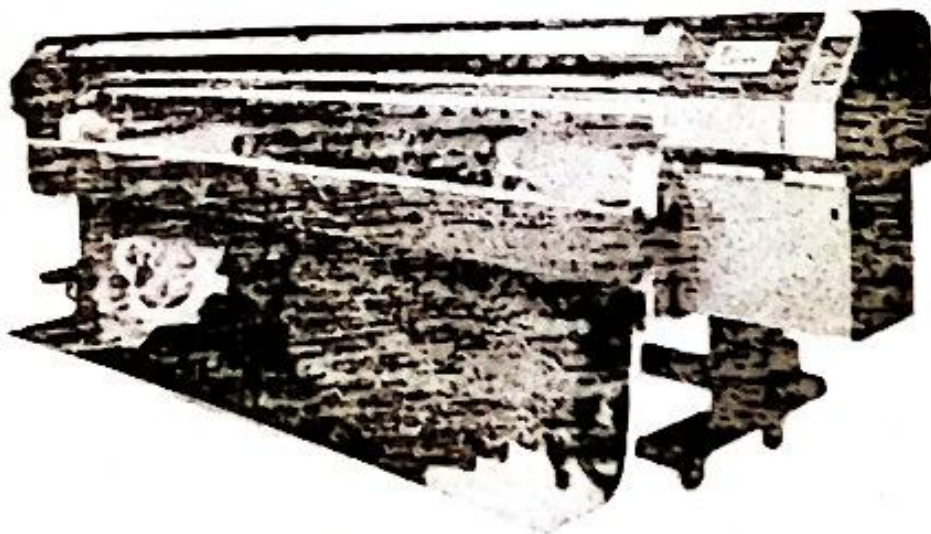


Fig. Ink Jet Printer

### 3. Thermal Printer:

Thermal printer uses color waxes and heat to produce images on special paper. The color wax sheets are not required for black and white output. Thermal printers are expensive and require costly paper. These printers are used to create the highest quality color printing.



Fig. Thermal Printer

**Q: Difference between Impact and Non-Impact Printer:**

**Ans.**

| IMPACT PRINTER                                                                                                       | NON IMPACT PRINTER                                                                                  |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Impact Printers produces characters and graphics on a piece of paper by <b>striking</b> it is called impact printer. | A type of printer that produces characters and graphics on a piece of paper <b>without striking</b> |
| It prints by hammering a set of metal pin or character set.                                                          | Printing is done by depositing ink in any form.                                                     |
| Electromechanical devices are used                                                                                   | No electromechanical device is used                                                                 |
| Impact printers are normally <b>slow</b> .                                                                           | Non-impact Printers like LASER printers are <b>fast</b> .                                           |
| Impact Printers create <b>noise</b> by striking it on ribbon.                                                        | Non-impact printers <b>do not create noise</b> as they do not strike                                |
| Dot-matrix printer, Daisy wheel printers, line printer are examples.                                                 | Inkjet printers, Thermal printers, laser printers are examples.                                     |

**Q: What is Plotter? Discuss different types of plotters.**

**Ans.** "Plotters are printers that prints on large sheets. So we can say that they are large printers. The printing output is created in vector graphics which means the dots are printed which are joined by lines and curves to produce text and images." The plotters are available in various designs and primarily used for printing large format prints that include architecture layouts, building plans, ship designs, CAD (computer-aided design), big canvas prints and any other engineering drawing.

There are two types of plotters are following:

- **Flatbed Plotter**
- **Drum Plotter**

#### Flatbed Plotters:

In **flatbed plotters**, the writing head moves in both x and y-axis directions across the surface of the paper. This type of plotter consists of a flat surface on which the paper is placed in a stationary position and it is the pen that moves to create drawing or images. The pen can be of different sizes and colors in flatbed plotters.



**Fig. Flatbed Plotters**

Flatbed Plotters need space where the bed reside. It can print according to the size of the bed. In order to create the big size print, we need to increase the size of the bed.

#### **Drum Plotters:**

There are other types of plotters called **drum plotters** in which drum or a cylindrical surface rotates in the vertical direction while the pen moves in another axis to create an image or other output. Moreover, the width of the final print-out is limited by the width of the drum while there is **no limit to the length** of the paper.



**Fig. Drum Plotter**

**Q:** Discuss basic units of storage.

**Ans.** Basic units of storage are follows:

**Bit:**

A "**bit**" is the smallest unit of storage. A bit stores just a 0 or 1. Computer is digital so it stores Binary which is 0 and 1.

**Byte:**

A byte consists of 8 bits. It is data measurement unit. A single byte can represent total  $2^8$  or 256 values (0 to 255). Normally a Byte is used to store a single character in memory. E.g. "F" is stored in computer as "01000110".

**Kilo Byte (KB):**

A collection of  $2^{10}$  or 1024 bytes is called Kilobyte. It is denoted as **KB**.

$$1 \text{ KB} = 2^{10} \text{ Bytes} = 1024 \text{ Bytes}$$

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**Megabyte (MB):**

A collection of  $2^{10}$  or 1024 Kilobytes is called Megabyte. It is denoted as MB

$$1 \text{ MB} = 2^{10} \text{ KB}$$

$$1 \text{ MB} = 2^{10} \times 2^{10} = 2^{20} \text{ Bytes} = \text{about one million bytes}$$

**Gigabyte (GB):**

A collection of  $2^{10}$  or 1024 Megabytes is called Gigabytes. It is denoted as GB.

$$1 \text{ GB} = 2^{10} \text{ MB}$$

$$1 \text{ GB} = 2^{10} \times 2^{10} = 2^{20} \text{ KB}$$

$$1 \text{ GB} = 2^{20} \times 2^{10} = 2^{30} \text{ Bytes} = \text{about one billion bytes}$$

**Terabyte (TB):**

A collection  $2^{10}$  or 1024 Gigabytes is called Terabyte. It is denoted as TB.

$$1 \text{ TB} = 2^{10} \text{ GB}$$

$$1 \text{ TB} = 2^{10} \times 2^{10} = 2^{20} \text{ MB}$$

$$1 \text{ TB} = 2^{20} \times 2^{10} = 2^{30} \text{ KB}$$

$$1 \text{ TB} = 2^{30} \times 2^{10} = 2^{40} \text{ Bytes} = \text{about one trillion bytes}$$

**Word:**

A word is a number of bits that constitute a common unit of data. It is normally the size of register. The length of words gets changed from computer to computer. Large size of a word means the more powerful computer.

A collection of 4 Bytes or 32 bits is called a word used in mainframe, mini-computer and microcomputer. 64 bits double word is used in some super computers.

$$1 \text{ Word} = 4 \text{ Bytes} = 32 \text{ Bits}$$

$$1 \text{ Double Word} = 8 \text{ byte} = 64 \text{ bits}$$

**Q: Discuss storage devices in detail.**

**Ans.** Storage devices are the computer hardware used to remember/ store data. A storage device may also be known as a storage medium or storage media.

Storage devices are available in different forms, depending on the type of underlying device. For example, a standard computer has multiple storage devices including RAM, a cache, and a hard disk. The same device may also have optical disk drives and externally connected USB drives. There are two different types of storage devices:

**Primary storage devices:** Generally smaller in size, primary storage devices are designed to hold data temporarily and are internal to the computer. They have the fastest data access speed. These types of devices include RAM and cache memory.

**Secondary storage devices:** Secondary storage devices usually have larger storage capacity, and they store data permanently. They can be either internal or external to the computer. These types of devices include the hard disk, the optical disk drive and USB storage device.

**Types of secondary storage devices:**

### III. Solid State Storage Devices

Solid-state storage (SSS) is a type of non-volatile computer storage that stores and retrieves digital information using only electronic circuits, without any involvement of moving mechanical parts.

#### (i) Solid State Drive

Solid-state drive (SSD) is a storage medium that uses non-volatile memory to hold and access data. Unlike a hard drive, an SSD has no moving parts, which gives it advantages such as faster access time, noiseless operation, higher reliability, and lower power consumption.

#### (ii) USB Flash Drive

A USB flash drive is a device used for data storage that includes a flash memory and an integrated Universal Serial Bus (USB) interface. Most USB flash drives are removable and rewritable. Physically, they are small, durable and reliable. The larger their storage space, the faster they tend to operate.

#### (iii) Memory Card

A memory card is a type of storage device that is used for storing media and data files. It provides a permanent and non-volatile medium to store data and files from the attached device. Memory cards are commonly used in small, portable devices, such as cameras and phones. A memory card is also known as a flash card.

#### (iv) Cloud Storage

Cloud storage is also known as online storage. It is an internet service that provides remote storage to users. Many service providers are there who provide online access to storage media. They offer limited services free of cost and premium services with financial charges.

**Cloud storage** is a service that allows you to **store data (files, photos, videos, documents)** on the internet instead of your computer's hard drive. These files are saved on remote servers and can be accessed from **any device with an internet connection**.

These are the examples of cloud storage:

##### ☞ 1. Google Drive

- **Provider:** Google
- **Storage:** 15 GB free (shared with Gmail and Google Photos)
- **Features:**
  - Create and store documents, spreadsheets, presentations using Google Docs.
  - Easy sharing and collaboration.
  - Syncs across devices using Google Drive app.
- **Usage:** Ideal for Google account users and collaboration with Google Workspace.

##### ☞ 2. Dropbox

- **Provider:** Dropbox Inc.
- **Storage:** 2 GB free (more with referrals or paid plans)

- **Features:**
  - Simple and clean interface.
  - Strong file synchronization and sharing features.
  - Supports file versioning and recovery.
- **Usage:** Popular for business use and file backup.
- △ **3. SkyDrive (now OneDrive)**
  - **Provider:** Microsoft
  - **Storage:** 5 GB free (more with Microsoft 365)
  - **Features:**
    - Integrated with Microsoft Office (Word, Excel, PowerPoint).
    - File syncing across devices.
    - Secure and easy to use with Windows systems.
  - **Usage:** Great for users of Windows and Microsoft Office tools.
- △ **4. iCloud**
  - **Provider:** Apple
  - **Storage:** 5 GB free (expandable with subscription)
  - **Features:**
    - Seamless syncing with iPhone, iPad, and Mac.
    - Stores photos, videos, contacts, calendars, and documents.
    - Works best in the Apple ecosystem.
  - **Usage:** Ideal for Apple device users.

#### Common Benefits of Cloud Storage Devices

- **Access files anytime, anywhere.**
- **Backup and data recovery.**
- **Easy sharing and collaboration.**
- **Secure storage with encryption.**

### Short Questions

1. **What is data?**

**Ans.** A collection of raw facts and figures related to an object is called data. An object may be a person, an organization, or any other thing etc. Data is collected for a specific purpose. Data may be in the form of text, numbers, images, sounds, and videos.

2. **Define information.**

**Ans.** The processed data is called information. The processed data gives us a clear and proper meaning. So it can be used for decision making.

3. **Differentiate between data and information.**

**Ans.** Data is raw facts and figures, while information is processed data. Data cannot be used for making decision, while information can be used for decision making. Data is used as input to the computer, while information is the output of the computer.

4. **Define data processing?**

Ans. The process of converting data into information is called data processing. In data processing, data is used as input and information is the output.

5. **What is information technology?**

Ans. The technology used for processing, storing and communicating information is called information technology or IT. It is the combination of hardware and software. It combines computing with high speed computer networks. Website is a type of information technology.

6. **What is global village?**

Ans. Global village means people living in the world know each other as if they are in a village. People can exchange information or messages in such a manner as if they are sitting in a same room.

7. **What is digital convergence?**

Ans. To convert in a process or organization work into digital form is called digital convergence. Now a days, most of the work is digitalized (computerized) now.

8. **State some application of information technology.**

Ans. Applications of information technology are stated below:

- Artificial Intelligence
- Web-Based Applications
- E-commerce
- M-commerce
- Computer animation
- Multimedia
- Hypermedia
- Distributed computing etc

9. **What is artificial intelligence?**

Ans. Artificial intelligence is intelligence exhibited by machines. These machines can think, make decision and communicate with human beings.

10. **What is web-based applications?**

Ans. A web-based application is any program that is accessed over a network connection using HTTP, rather than existing within a device's memory. Web-based applications often run inside a web browser.

11. **What is E-commerce?**

Ans. E-commerce is doing business transactions through computer networks, i.e. Internet.

12. **What is M-commerce?**

Ans. M-commerce is doing business transactions through mobile on the Internet.

13. **What is computer animation?**

Ans. The animation of moving pictures and cartoons.

14. **What is multimedia?**

Ans. Multimedia is the combination of text, audio, video, and graphics.

15. **What is hypermedia?**

Ans. Hypermedia is a type of multimedia that includes links to other information.

16. **What is a computer hardware?**

Ans. A computer hardware is a physical component of a computer system.

17. **What is a software?**

Ans. The physical components of a computer system that are not hardware.

18. **What is a set of software?**

Ans. A set of software is a collection of programs that perform a specific task.

19. **What is a program?**

Ans. A program is a set of instructions that tell a computer what to do.

20. **What is a problem?**

Ans. A problem is a situation that requires a solution.

21. **What is a solution?**

Ans. A solution is a way of solving a problem.

22. **What is a computer?**

Ans. A computer is a machine that can store and process data.

23. **What is a network?**

Ans. A network is a group of computers connected together.

**13. What is computer animation?**

**Ans.** The animation means cartoon or moving picture. Animation is a process of creating moving pictures and other special effects for television shows, commercials, movies and cartoons.

**14. What is multimedia?**

**Ans.** Multimedia is content that uses a combination of different content forms such as text, audio, images, animations, video and interactive content. Main use of multimedia applications are in video games and computer based training.

**15. What is hypermedia?**

**Ans.** Hypermedia is an extension of the term hypertext, is a nonlinear medium of information that includes graphics, audio, video, plain text and hyperlinks. Hypermedia is the process of creating links in multimedia files.

**16. What is a computer?**

**Ans.** A computer is an electronic machine. It can be programmed to accept data, process data into useful information, and store it for later use. A computer system consists of hardware and software.

**17. What is hardware?**

**Ans.** The physical components of a computer are called hardware. You can touch, see and feel hardware. The input/output devices, CPU, memory unit etc are examples of hardware.

**18. What is software?**

**Ans.** A set of instructions given to the computer to perform a specific task is called software. Software is also called computer program.

**19. What is system software?**

**Ans.** A program or set of programs that is especially designed to control different operations of computer system is called system software. It controls the working of different components of the computer. The system software also enables the other application programs to execute properly.

**20. What is application software?**

**Ans.** A program or set of programs that are especially designed to solve the specific problems of users are called application software. For exp, Microsoft Word, etc.

**21. What is the difference between application and system software?**

**Ans.** Application software solves the problem of user, while system software controls different operations of the computer. The application software depends on system software to run on the computer.

**22. What is the use of customized software?**

**Ans.** The software that is developed for a particular customer or organization is called customized built software. It is also known as customized software.

**23. What is packaged software?**

**Ans.** The software that is developed for sale to the general public is called packaged software. Packaged software is used to solve some common problem of many people.

or users. Examples of package software are spreadsheet and word processing programs.

**24. What is utility software?**

**Ans.** A software that helps the user to solve the specific problems related to hardware is called utility program. It is system software. A large number of programs are available for the service of hardware and software.

**25. What is operating software?**

**Ans.** An operating system is a collection of system programs that controls and coordinates the overall operations of a computer system. It provides and interfaces the user to interact with the computer. A computer needs an operating system to work. Examples of operating system are Microsoft Windows, DOS, Linux, Solaris, Android, IOS and Mac OS.

**26. What is the difference between hardware and software?**

**Ans.** Hardware are physical parts of the computer, while software is a set of instructions given to the computer to solve a problem. We can touch hardware but we can't touch software.

**27. What is input?**

**Ans.** Anything given to the computer is called input. The data and instructions are given to the computer as input. There are two methods of entering data and instructions to the computer. These are direct and indirect input.

**28. What is the difference between direct input and indirect input?**

**Ans.** Using direct input, data is entered directly into the computer from the source. In indirect method, data is entered into the computer after intermediate handling.

**29. What are input devices?**

**Ans.** The devices that are used to enter data and instructions into the computer are called input devices. The input device is a hardware component. It takes input from user and converts it into a form that a computer can use.

**30. State different categories of input devices.**

**Ans.** There are three general types of input devices. These are:

- Keyboards
- Pointing devices
- Source data-entry devices

**31. What is the function of keyboard?**

**Ans.** Keyboard is the most commonly used input device. It is mostly used to enter text and numeric type data into the computer. The buttons on the keyboard are called keys. A standard keyboard contains more than 105 keys.

**32. What is meant by QWERTY?**

**Ans.** The standard keyboard is also called QWERTY keyboard. This is because the first five leftmost keys on the top row of alphabets are Q, W, E, R, T, Y.

**33. What is a mouse?**

**Ans.** There are two types of mouse: wired and wireless. It is used to point and click on the screen.

**34. What is a touchpad?**

**Ans.** A touchpad is a pointing device that is used to control the cursor on the screen. It is commonly found on laptops and tablets.

**35. What is a trackball?**

**Ans.** A trackball is a pointing device that is used to control the cursor on the screen. It is commonly found on desktop computers and laptops.

**36. How does a scanner work?**

**Ans.** A scanner is a device that is used to capture images or documents from a physical source and convert them into a digital format. It can be used to scan documents, photos, and other physical objects.

**37. What is a digital camera?**

**Ans.** A digital camera is a device that is used to capture images or videos in a digital format. It can be used to take photos and videos, and the files can be stored on a memory card or connected to a computer.

**38. What is a digital voice recorder?**

**Ans.** A digital voice recorder is a device that is used to capture and store audio recordings in a digital format. It can be used to record voice messages, interviews, and other audio content.

**39. What is a digital scale?**

**Ans.** A digital scale is a device that is used to measure the weight of objects in a digital format. It can be used to measure the weight of food, ingredients, and other objects.

**40. What is a digital thermometer?**

**Ans.** A digital thermometer is a device that is used to measure temperature in a digital format. It can be used to measure the temperature of the body, the environment, and other objects.

**33. What are function keys?**

**Ans.** There are 12 function keys on the top of the keyboard that are labeled as F1, F2.....F12. These are used to perform special functions. The function of each function key depends upon the software being used on the computer.

**34. What are pointing devices?**

**Ans.** A pointing device is an input device. It is used to control movement of the pointer or cursor on the screen. It can also be used for sending command signals to the computer, selecting items on the screen, selecting commands from commands menu and drawing graphs etc. Examples of pointing devices includes Mouse, Trackball, Pointing stick, Joystick, Touch pad, Light pen, etc.

**35. What is the function of mouse?**

**Ans.** Mouse is the most commonly used pointing input device. It is used to control the cursor or pointer on the screen and to give commands to the computer. It is very easy to use. The mouse is attached with the computer by a cable or wireless connection.

**36. How does mouse work?**

**Ans.** Mouse is moved over a flat surface to control the movement of the pointer on the screen. The top of the mouse contains one wheel and 2 or 3 buttons. These buttons can be clicked or double clicked to perform different tasks. A mechanical mouse has a small ball at the bottom. The ball rotates over a flat surface. As the mouse is rolled over the flat surface, the pointer moves in the same direction on the screen.

**37. What is trackball?**

**Ans.** A trackball is a pointing input device. It performs functions like a mouse but it is a stationary device. It has a moveable ball on its top. The ball is rotated or rolled with fingers to move the pointer on the screen. The trackball is usually available with laptop computer. It is fixed on its keyboard. It is also available as separate device.

**38. What is touchpad or track pad?**

**Ans.** Touch pad is a pressure-sensitive pointing input device. Touch pad is also known as track pad. It is also stationary device like trackball but it has no moving parts. It is small, flat surface over which a user slides fingertip to move the pointer on screen. Touch pad is normally used with laptop.

**39. What is pointing stick?**

**Ans.** A pointing stick is a pressure-sensitive pointing device. It looks like a pencil eraser. It exists between keys of keyboard. It is used to control the movement of a pointer on the screen. The pointer on the screen moves in the direction in which the pointing stick is pushed. Pointing stick input device is normally used with laptop computers.

**40. What is touch screen?**

**Ans.** A touch screen is a special video display screen. Input is given to the computer directly by touching the screen with user fingertip. Most touch screen computers use sensors in the computer's screen to detect the touch of a finger. Touch screen is often used in ATM. Touch screen systems are also used in restaurants, supermarkets departmental stores etc.

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**41. What is joystick?**

**Ans.** Joystick is a pointing device. It consists of a vertical handle or handheld stick. The stick is mounted on a base. The handheld stick is used to control the movement of a pointer on the screen. Joystick is basically used to play video games. It is also used in some computer aided design (CAD) Systems.

**42. What are pen based systems?**

**Ans.** Pen based system use a pen like input device. This input device is called stylus. The system allows the user to enter handwriting and marks into computer. The user writes information on the display screen using stylus. The pen based system uses special software. This software translates handwritten information or marks into a form that can be stored and processed by the computer.

**43. What is graphic tablet?**

**Ans.** Digitizer is used for drawing new images or tracing old images. Usually, it is used for making maps and engineering drawings. A digitizer is also known as graphics tablet. It consists of a flat, rectangular board and an input device stylus or puck. Each location on the rectangular board points to corresponding position on the computer screen. The stylus or puck is connected to the board by a wire.

**44. What are source data entry devices?**

**Ans.** The devices that are used for direct data entry to the computer system are called source data entry devices. These devices are also known as direct input devices. These devices provide quick and efficient way to input data into the computer. The commonly used source data entry devices are image scanner, barcode reader, MICR, OMR, OCR, etc.

**45. What is impact printer?**

**Ans.** A type of printer produces output on a paper by striking the print hammer or wheel against an inked ribbon is called impact printer.

**46. What is non-impact printer?**

**Ans.** A printer that produces output on paper without striking the paper is called non-impact printer.

**47. What is barcode?**

**Ans.** Barcode is a unique identification code. It consists of light and dark vertical lines with spaces between them of different widths. Barcode is printed on most of the manufactured products. It represents the information about the product. The most popular and commonly used barcode is UPC.

**48. What is UPC?**

**Ans.** UPC stands for Universal Product Code. It is most popular and commonly used barcode system or scheme. It is printed on most of the manufactured products. It represents the information about the product.

**49. What is barcode reader?**

**Ans.** Barcode reader is a special scanning device. It is used for reading barcode printed on the product or book. It used laser beam to scan or read the barcode. It reads the barcode and translates it into digital signals. These signals are input to the computer for further processing.

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**50. What are mark and character recognition devices?**

**Ans.** Mark and character recognition devices are input devices. They are used to scan information printed on the paper. These devices read the information and convert it into digital signals. These signals are input to the computer for further processing. Examples of these devices are MICR, OMR, OCR etc.

**51. What is MICR?**

**Ans.** MICR stands for Magnetic-Ink Character Recognition. It is a special scanner. It reads the characters printed with magnetic ink and converts them into digital signals. These signals are then input to the computer for further processing. MICR devices are normally used in banks for processing cheques.

**52. What is OMR?**

**Ans.** OMR stands for Optical Marks Recognition. OMR device is used to read handwritten marks or symbols printed on the paper. It uses a light beam to scan the marks and converts them into digital signals. These signals are then input to the computer for further processing. OMR device is usually used in education field to check objective type answers.

**53. What is OCR?**

**Ans.** OCR stands for Optical Character Recognition. OCR device reads preprinted characters in a particular font and converts them into digital code. The characters printed on paper may be typewritten or handwritten. The OCR devices are commonly used to process utility bills and price code printed on the products in departments stores.

**54. What is magnetic strip card?**

**Ans.** A magnetic strip card has a strip of magnetically encoded information on its back. Usually, the magnetic strip is used in personal identity card. This card is used for personal identification during driving, in the departmental stores, and at public places etc.

**55. What is smart card?**

**Ans.** Smart card looks like a credit card or ATM card. This card contains a thin microprocessor and memory chip that is embedded in it. The card is inserted into a card reader. The card reader reads information stored on the card. It can also store and update some basic information. A mobile SIM and an ATM card are examples to smart card.

**56. What is facsimile?**

**Ans.** A fax machine is also known as facsimile transmission machine. It is used to send and receive information printed on paper through telephone line.

**57. What is image scanner?**

**Ans.** Image scanner is also called graphic scanner. It reads text, drawing and photographs printed on the paper. It converts them into digital form and stores inside the computer for further processing.

**58. What is raster graphics?**

**Ans.** The technique of representing a graphic image as a matrix of dots is called raster graphics.

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**59. What is audio input?**

**Ans.** Audio input is the process of entering any sounds into the computer. It may include speech, and music etc. The sound is in the form of analog signals.

**60. What is audio input devices?**

**Ans.** The device that is used to enter sound into the computer is called audio input device. Microphone is most commonly used as an audio input device.

**61. What is video input?**

**Ans.** The process of entering video input into the computer is called video input. Video input consists of full motion images that create actions and movements. The video input is also in the form of analog signals.

**62. What are video input devices?**

**Ans.** The devices that are used to enter video input into the computer are called video input devices. Most of these devices are used for recording movie films and video images. VCR and video camera are examples of video input devices.

**63. What is digital camera?**

**Ans.** A digital camera is an input device. It takes pictures and stores them in digital form. It does not use traditional camera film. It uses a light sensitive processor chip to capture photographic images in digital form. These images are stored on a small diskette inserted in the camera or on flash memory chip.

**64. What is output?**

**Ans.** The data processed into a useful form is called output. The user receives output from the computer on a screen or print on the paper through printer. In case of audio output, then it can be received through speaker.

**65. What is softcopy output?**

**Ans.** The output received on the display screen is called softcopy output. The output in the form of audio or video or file stored on disk is also softcopy output. This kind of output is not tangible. It means that softcopy output cannot be touched.

**66. What is hardcopy output?**

**Ans.** The output printed on the paper is called hardcopy. Hardcopy can be touched, while softcopy cannot be touched. Hardcopy is received through printer or plotter, while softcopy is received on display screen or storage device or speaker.

**67. What are output devices?**

**Ans.** A hardware that is used to receive the output from the computer is called an output device. It takes information from the computer and converts it in a form that a user can understand. The commonly used output devices are monitors, printers, plotters and speakers.

**68. What is display screen?**

**Ans.** Display screen is also called monitor or simply a screen. It is commonly used output device. It is used as softcopy output device. The output may be in the form of text, graphics or video.

## ILMI Applications

**69. What is**

**Ans.** Pixel stands for picture element. Dots are made up of pixels. The number of pixels in a picture is called resolution.

**70. Why re**

**Ans.** Resolution is the number of pixels in a picture. The factor that affects the clarity of the picture is resolution.

**71. What i**

**Ans.** A display device is used to show the output of the computer as video or text. It can be a monitor or a printer.

**72. What**

**Ans.** Laser printer is a type of printer that uses a laser beam to create the image on the paper. The laser beam is focused on the paper and the toner is applied to the paper.

**73. What**

**Ans.** Inkjet printer is a type of printer that uses ink to create the image on the paper. The ink is sprayed onto the paper and the toner is applied to the paper.

**74. What**

**Ans.** The speed of the printer is measured in terms of pages per minute (ppm). The higher the ppm, the faster the printer.

**75. Wh**

**Ans.** The resolution of the printer is the number of dots per inch (dpi). The higher the dpi, the sharper the image.

**76. Bit**

**Ans.** A bit is the smallest unit of data in a computer. It can be either 0 or 1. A byte is made up of 8 bits.

**77. Cor**

**Ans.** A corrupt file is a file that has been damaged or altered. It may not be able to be opened or used.

**78. Bit**

**Ans.** A bit is the smallest unit of data in a computer. It can be either 0 or 1. A byte is made up of 8 bits.

69. **What is pixel?**

**Ans.** Pixel stands for picture element. The display screen is divided into small dots. These dots are called pixels. The pixels are arranged horizontally and vertically. The number of pixels on a screen is called its resolution. The monitor's screen that has large number of pixels has high resolution.

70. **Why resolution is important for display screen?**

**Ans.** Resolution is very important factor for display screen. The resolution is the main factor that affects the quality of the monitor. The monitor's screen that has large number of pixels has high resolution. The resolution refers to the sharpness and clearness of an image.

71. **What is video display adapter?**

**Ans.** A display screen must have a video display adapter attached with the computer. It is used to display output in the form of graphics. Video display adapter is also known as video graphics card. It is a circuit board that determines the resolution, number of colors and speed with which images appear on the display screen.

72. **What is laser printer?**

**Ans.** LASER stands for Light Amplification by Stimulated Emission of Radiation. A laser printer is the fastest and high quality non-impact printer. It works like a photocopier. The laser printer transfers the image of output on paper using LASER technology and toner. Toner is an ink powder.

73. **What is inkjet printer?**

**Ans.** Inkjet printer is a type of non-impact printer. It creates output on paper by spraying tiny droplets of liquid ink. Inkjet printer has a print-head that can spray very fine drops of ink. It consists of print cartridge filled liquid ink and has small nozzles in the form of matrix. The combination of nozzles is activated to form the shape of character or image on the paper by spraying the liquid ink.

74. **What is thermal printer?**

**Ans.** Thermal printer is another type of non-impact printer. It can only print output on a special heat sensitive waxy paper. The image of the output is created on the waxy paper by burning dots on it. For coloured output, coloured waxy sheets are used.

75. **What are basic units of data storage?**

**Ans.** The basic units that are used for data storage are as follow:

- Bit:** The binary digit 1 or 0 is called bit. It is the basic unit for storing data in the computer memory.
- **Byte:** A group of 8 bits is called byte. One character takes one byte of memory.
  - **Kilobyte:** A kilobyte consists of 1024 Bytes. It is denoted as KB.
  - **Megabyte:** A Megabyte consists of 1024 Kilobytes. It is denoted as MB.
  - **Gigabyte:** A Gigabyte consists of 1024 Megabytes. It is denoted as GB.
  - **Terabyte:** A Terabyte consists of 1024 Gigabytes. It is denoted as TB.

